AI-DRIVEN TRANSFORMATION IN BANKING GRC: IMPLICATIONS FOR WORKFORCE MANAGEMENT IN UK AND EU BANKS

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ABSTRACT

Banks in the United Kingdom (UK) and the European Union (EU) are adopting Artificial Intelligence (AI) across their Governance, Risk and Compliance (GRC) functions. The return on this investment depends less on the technology than on the workforce that must operate, supervise and answer for it. At the same time, the EU Digital Operational Resilience Act (DORA), the EU Artificial Intelligence Act, PRA Supervisory Statement SS1/23 and the UK Senior Managers and Certification Regime (SM&CR) turn staff competence and human oversight into regulatory obligations. Existing research covers RegTech, workplace automation and technology acceptance separately, but not their combined effect on workforce management in regulated banking GRC.

The study examines how AI-driven transformation in UK and EU banking GRC is reshaping workforce management decisions on role structure, skill requirements, staffing and employee adaptation, and develops a practical framework for GRC and HR leadership. It follows a sequential exploratory mixed-methods design. Written semi-structured interviews with GRC practitioners and senior professionals produced ten themes. A structured questionnaire, built on constructs adapted from Davis (1989), Glikson and Woolley (2020) and Brougham and Haar (2018), measured how strongly practitioners agree with these themes, reported as means and standard deviations. The themes were then translated into hypotheses that were tested statistically on four independent public datasets: Eurostat enterprise AI-adoption data for 27 EU

countries, the annual reports of large UK and EU banks for FY2022 and FY2025, UK financial-services job adverts, and published industry surveys. Parametric and non-parametric tests were applied, and all evidence was triangulated against four research propositions.

The findings show that routine, rules-based GRC work is automated first, while judgement-intensive work is augmented rather than replaced. AI adoption among large EU enterprises roughly doubled between 2021 and 2025, and AI references in bank annual reports rose about tenfold between FY2022 and FY2025, rising in every paired bank. References to DORA, the AI Act and SS1/23 and the share of banks naming a dedicated AI governance body also rose significantly. AI-exposed job adverts demand markedly more data, automation and cyber skills. Reskilling disclosure, however, grew far more slowly than AI disclosure and was not linked to AI intensity. Practitioners rated training adequacy lowest among skills items and identified the loss of institutional knowledge, and a weak understanding of AI among accountable senior managers, as the main hidden risks.

The study concludes that AI in GRC recomposes roles rather than removing them, that regulation makes competence and oversight auditable, and that banks' workforce response lags their AI ambition. It proposes a seven-stage workforce transformation framework: diagnose, protect knowledge, redesign roles, build oversight competence, reskill and source, secure trust, and lead change. Knowledge protection precedes any restructuring. The framework gives GRC and HR leaders a sequenced, regulation-aware plan for managing the transition.

(Signature of the Candidate)

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(Signature)

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TABLE OF CONTENTS

PAGE NO.

ABSTRACT.....iv

LIST OF TABLES.....xi

LIST OF FIGURES.....xiii

LIST OF ABBREVIATIONS.....xiv

CHAPTER 1 - INTRODUCTION.....1

1.1 RESEARCH BACKGROUND.....1

1.2 IDENTIFIED PROBLEM.....2

1.3 NEED FOR THE STUDY.....3

1.4 OBJECTIVES OF THE STUDY.....4

1.4.1 Primary objective.....4

1.4.2 Secondary objectives.....4

1.5 SCOPE OF THE STUDY.....4

1.6 DELIVERABLES.....5

1.7 ORGANISATION OF THE REPORT.....6

CHAPTER 2 - LITERATURE SURVEY.....7

2.1 INTRODUCTION.....7

2.2 TECHNOLOGY ACCEPTANCE AND TRUST IN AI.....7

2.2.1 Technology Acceptance Model (Davis, 1989).....7

2.2.2 Human trust in AI (Glikson and Woolley, 2020).....8

2.3 AUTOMATION AND THE RECOMPOSITION OF WORK.....8

2.3.1 Substitution and complementarity (Autor, 2015).....8

2.3.2 Susceptibility to computerisation (Frey and Osborne, 2017).....9

2.3.3 Technology, work and organisations (Cascio and Montealegre, 2016)….9

2.4 REGTECH AND THE TRANSFORMATION OF COMPLIANCE.....10

2.4.1 The concept of RegTech (Arner, Barberis and Buckley, 2017).....10
2.4.2 The European route to RegTech (Buckley, Arner, Zetzsche and Weber, 2020).....11
2.5 EMPLOYEE PERCEPTIONS OF AI AND AUTOMATION.....11
2.5.1 STARA awareness (Brougham and Haar, 2018).....11
2.6 REGULATORY CONTEXT: WORKFORCE CAPABILITY AS AN OBLIGATION.....12
2.7 SUMMARY OF THE LITERATURE REVIEWED.....13
2.8 RESEARCH GAP.....13
2.9 LINKING THE LITERATURE TO THE STUDY.....14
CHAPTER 3 - METHODOLOGY.....16
3.1 TYPE OF PROJECT.....16
3.2 DATA REQUIREMENT.....17
3.3 TARGET RESPONDENTS.....17
3.4 SAMPLING METHODS.....18
3.4.1 Primary data.....18
3.4.2 Secondary data used for statistical testing.....18
3.5 RESEARCH INSTRUMENTS.....20
3.5.1 Semi-structured interview guide.....20
3.5.2 Structured questionnaire.....20
3.5.3 Method of preparing the instruments.....20
3.6 RESEARCH PROPOSITIONS AND HYPOTHESES.....21
3.6.1 Research propositions.....21
3.6.2 Questionnaire hypotheses (descriptive).....21
3.6.3 From interview themes to public-data hypotheses.....22
3.6.4 Statistical hypotheses.....23
3.7 DATA PROCESSING.....24
3.8 TOOLS FOR ANALYSIS.....25
3.9 ASSUMPTIONS, CONSTRAINTS AND LIMITATIONS.....26
3.10 ETHICAL CONSIDERATIONS.....28
CHAPTER 4 - DATA ANALYSIS AND INTERPRETATION.....29
4.1 INTRODUCTION.....29

4.2 PROFILE OF RESPONDENTS.....29
4.3 THEMATIC ANALYSIS OF INTERVIEW DATA.....30
4.3.1 Objective (i): activities automated or augmented, and role demand.....30
4.3.2 Objective (ii): regulation, human oversight and competence.....31
4.3.3 Objective (iii): skill gaps and reskilling.....32
4.3.4 Objective (iv): employee behavioural responses.....33
4.4 ANALYSIS OF QUESTIONNAIRE DATA (DESCRIPTIVE STATISTICS) 33
4.5 ANALYSIS OF SECONDARY DATA.....36
4.5.1 Extent of AI adoption and the skills constraint (Objective i).....36
4.5.2 Regulatory requirements mapped to workforce competence and oversight (Objective ii).....37
4.5.3 Changing job descriptions and new roles (Objectives i and iii).....38
4.5.4 Workforce management practices (Objectives iii and iv).....40
4.6 STATISTICAL TESTING OF HYPOTHESES ON PUBLIC DATA.....41
4.6.1 AI adoption across EU economies (H5, dataset D1).....41
4.6.2 Bank annual reports (H10, dataset D2).....42
4.6.3 Job adverts (H6 and H7, dataset D3).....44
4.6.4 Published surveys (H8 and H9, dataset D4).....45
4.6.5 Summary of hypothesis results.....46
4.7 TRIANGULATION OF EVIDENCE.....46
4.7.1 Practitioner views and industry data.....46
4.7.2 Testing of propositions.....47
4.8 GRC ROLE-IMPACT MAP.....48
4.9 WORKFORCE TRANSFORMATION FRAMEWORK.....49
4.10 DELIVERABLES.....50
CHAPTER 5 - CONCLUSIONS.....52
5.1 SUMMARY OF FINDINGS.....52
5.2 SUGGESTIONS AND RECOMMENDATIONS.....54
5.2.1 For GRC leadership.....54
5.2.2 For HR leadership.....55
5.2.3 For boards and accountable senior managers.....55
5.3 CONCLUSIONS.....56

5.4 DIRECTIONS FOR FUTURE RESEARCH.....57

APPENDIX 1 - WRITTEN INTERVIEW: PRACTITIONER RESPONSE FORM 58

APPENDIX 2 - STRUCTURED QUESTIONNAIRE.....61

APPENDIX 3 - BANK ANNUAL REPORT SAMPLE AND CODING DICTIONARY.....64

A3.1 Banks in the D2 sample.....64

A3.2 D2 coding dictionary.....64

A3.3 D3 skill dictionary.....65

REFERENCES.....66

LIST OF TABLES

PAGE NO.

Table 2.1 Summary of the literature reviewed.....13
Table 2.2 Link between literature, objectives and propositions.....14
Table 3.1 Data requirement, sources and objectives served.....17
Table 3.2 Public datasets used for hypothesis testing.....18
Table 3.3 Questionnaire constructs.....20
Table 3.4 Mapping of instruments to objectives.....21
Table 3.5 Research propositions.....21
Table 3.6 Translation of interview themes into public-data hypotheses.....22
Table 3.7 Statistical hypotheses tested on public data.....23
Table 3.8 Variables used in the statistical analysis.....25
Table 4.1 Profile of respondents.....29
Table 4.2 Interview themes for Objective (i).....30
Table 4.3 Interview themes for Objective (ii).....31
Table 4.4 Interview themes for Objective (iii).....32
Table 4.5 Interview themes for Objective (iv).....33
Table 4.6 Item-level descriptive statistics (scale 1–5).....33
Table 4.7 Construct composite scores.....34
Table 4.8 Assessment of questionnaire hypotheses H1–H4.....35
Table 4.9 AI adoption and skills indicators in UK financial services.....36
Table 4.10 Regulatory requirements and their workforce implications.....37
Table 4.11 Changing emphasis in GRC job descriptions.....38
Table 4.12 New AI risk and compliance role families: share of public postings.....39
Table 4.13 Skills coded from public role profiles.....40
Table 4.14 Evidence on workforce management practices.....40
Table 4.15 Mean AI adoption and ICT-specialist share across 27 EU countries.....41
Table 4.16 Test results for H5.....41

Table 4.17 Mean (median) term density per 10,000 words, 19 paired banks.....42
Table 4.18 Test results for H10.....42
Table 4.19 Skill demand in AI-exposed and other adverts (H6).....44
Table 4.20 Advertised annual salary by role group (H7).....44
Table 4.21 Test results for H8 and H9.....45
Table 4.22 Summary of hypothesis results.....46
Table 4.23 Convergence of practitioner views and public data.....47
Table 4.24 Triangulated assessment of research propositions.....47
Table 4.25 GRC role-impact map.....48
Table 4.26 Seven-stage GRC workforce transformation framework.....49
Table 4.27 Deliverables of the study.....50
Table A 3.1 Banks in the D2 sample.....64
Table A 3.2 D2 coding dictionary.....65
Table A 3.3 D3 skill flags.....65

LIST OF FIGURES

PAGE NO.

Figure 4.1 AI adoption over time and its relationship with ICT-specialist share (EU27, Eurostat).....42
Figure 4.2 Change in AI, reskilling, AI-governance and regulation term density in bank annual reports, FY2022 to FY2025.....43
Figure 4.3 Skill demand and advertised salary in AI-exposed and other GRC adverts. .45

LIST OF ABBREVIATIONS

AI	Artificial Intelligence
AML	Anti-Money Laundering
AMLA	EU Anti-Money Laundering Authority
AMLR	EU Anti-Money Laundering Regulation
CDD	Customer Due Diligence
DORA	Digital Operational Resilience Act (Regulation (EU) 2022/2554)
EBA	European Banking Authority
ECB	European Central Bank
EU	European Union
FCA	Financial Conduct Authority
FS	Financial Services
FY	Financial Year
GenAI	Generative Artificial Intelligence
GRC	Governance, Risk and Compliance
HR	Human Resources
ICT	Information and Communication Technology
IIA	Institute of Internal Auditors
KYC	Know Your Customer
LLM	Large Language Model
MI	Management Information
ML	Machine Learning
OLS	Ordinary Least Squares
PRA	Prudential Regulation Authority
RegTech	Regulatory Technology
SD	Standard Deviation
SM&CR	Senior Managers and Certification Regime
SMF	Senior Management Function

SS1/23	PRA Supervisory Statement 1/23 on model risk management
STARA	Smart Technology, Artificial Intelligence, Robotics and Algorithms
TAM	Technology Acceptance Model
UK	United Kingdom

CHAPTER 1

INTRODUCTION

1.1 RESEARCH BACKGROUND

Banks in the United Kingdom (UK) and the European Union (EU) are investing substantially in Artificial Intelligence (AI) across their Governance, Risk and Compliance (GRC) functions. GRC covers the activities through which a bank governs itself, identifies and controls risk, and demonstrates compliance with law and regulation. It includes compliance monitoring and advisory, financial crime and anti-money laundering (AML) operations, know-your-customer (KYC) and customer due diligence (CDD), operational and ICT risk, model risk and validation, regulatory reporting and internal audit. These functions are labour-intensive, data-heavy and governed by detailed rules, which makes them natural candidates for automation and AI.

AI is now applied to regulatory monitoring, customer and transaction screening, risk scoring, control testing, document analysis and compliance reporting. The joint Bank of England and Financial Conduct Authority (FCA) survey of AI in UK financial services found that 75% of firms already use AI, with a further 10% planning to do so within three years (Bank of England & FCA, 2024). One third of AI use cases are third-party implementations, and risk and compliance is among the business areas with the highest share of such implementations, at 64%. The same survey found that 46% of firms have only a partial understanding of the AI technologies they use, and that insufficient talent and access to skills is a large or medium constraint for more than half of firms. Adoption is therefore moving faster than the capability needed to govern it.

Generative AI and large language models have widened the range of GRC tasks that can be automated or assisted. Earlier rules-based automation handled structured, repeatable work such as data aggregation and reconciliation. Newer tools can

draft case summaries, map regulatory obligations and review documents. This brings AI closer to work that was previously considered judgement-based and raises new questions about who checks AI output and who is accountable for it.

The regulatory environment makes these questions urgent. The EU Digital Operational Resilience Act (DORA) requires the management body to maintain sufficient knowledge of ICT risk and requires ICT security and resilience training for all staff. The EU Artificial Intelligence Act introduces an AI-literacy duty and requires effective human oversight of high-risk AI systems. In the UK, PRA Supervisory Statement SS1/23 sets expectations for model risk management, including the skills of those who develop and validate models, and the Senior Managers and Certification Regime (SM&CR) holds named individuals personally accountable for decisions that are increasingly AI-assisted. Staff competence and effective human oversight have therefore become regulatory conditions, not merely HR objectives.

1.2 IDENTIFIED PROBLEM

The business case for AI in GRC is usually framed around cost efficiency and faster regulatory response. In practice, the return depends less on the capability of the technology than on the readiness of the workforce that must operate alongside it. When AI absorbs routine monitoring and reporting work, management must decide:

- which GRC roles are redesigned, reduced or newly created;
- which skills must be built internally, bought in the market or borrowed from vendors;
- how teams should be structured and staffed; and
- how employees can be brought to trust and use AI-enabled controls appropriately rather than over-relying on them or working around them.

These are workforce management decisions, and at present they are taken with little empirical guidance specific to regulated banking. A bank that automates a compliance process without ensuring that its people can understand and challenge the output has not reduced its risk; it has relocated it. The problem addressed by this study is therefore a management problem rather than a technical one: how GRC and HR

leaders in UK and EU banks should plan role design, reskilling and employee adaptation as AI is introduced, while keeping compliance effective and employees engaged.

1.3 NEED FOR THE STUDY

The study is needed for four reasons.

First, the scale of change is significant and immediate. AI adoption in UK financial services is already widespread, and publicly advertised AI risk and compliance roles grew by 187% year-on-year while general technology hiring contracted, with financial services and banking among the leading hiring sectors (Global Skill Development Council, 2026).

Second, regulation now requires banks to demonstrate that their people are competent to supervise AI. Under DORA, the AI Act, SS1/23 and SM&CR, failures of competence or oversight become regulatory findings and, in the UK, personal exposures for senior managers.

Third, existing research does not address this intersection. The RegTech literature explains how technology is reshaping compliance but treats the workforce only implicitly. The work and employment literature explains how automation recomposes jobs, but across industries rather than within regulated banking functions. The technology acceptance and trust literature explains individual adoption largely through experimental rather than high-stakes settings.

Fourth, GRC and HR leaders need practical guidance. A sequenced, evidence-based framework linking AI adoption to role redesign, reskilling and employee adaptation would help banks avoid two common failures: cutting experienced staff before their knowledge is captured, and deploying AI faster than the organisation can oversee it.

1.4 OBJECTIVES OF THE STUDY

1.4.1 Primary objective

To examine how AI-driven transformation in the Governance, Risk and Compliance functions of UK and EU banks is reshaping workforce management decisions on role structure, skill requirements, staffing and employee adaptation, and to develop a practical framework enabling banking management to plan GRC workforce transformation while sustaining compliance effectiveness and employee engagement.

1.4.2 Secondary objectives

i. To map which GRC activities are being automated or augmented by AI in UK and EU banks, and how this is changing the demand for compliance and risk roles.

ii. To analyse how UK and EU regulatory requirements (DORA, the EU AI Act, SS1/23 and SM&CR) shape role design, human oversight and competence obligations within GRC functions.

iii. To identify emerging skill gaps and reskilling strategies in GRC functions, and to assess how they are being addressed by management.

iv. To examine employee behavioural responses to AI adoption in GRC (perceived usefulness, trust, perceived replaceability and readiness to adapt) and their implications for adoption and retention.

v. To develop an integrated workforce transformation framework and recommendations for GRC and HR leadership in UK and EU banks.

1.5 SCOPE OF THE STUDY

The study covers GRC functions in banks and banking service providers, including global capability centres, operating in the UK and the EU. Within GRC it covers compliance, risk, financial crime, model risk, operational and ICT risk, regulatory reporting and internal audit, together with the GRC technology and HR roles that support them.

The regulatory scope covers DORA, the EU AI Act as amended by the Digital Omnibus on AI (2026), PRA SS1/23 and SM&CR. Related instruments that bear directly on GRC competence, such as the FCA Consumer Duty, the EU Anti-Money Laundering Regulation and the ECB supervisory priorities for 2026–28, are considered where relevant.

The time frame for secondary data is 2021 to 2025, covering the period in which generative AI reached the market and DORA and the AI Act were adopted. Bank disclosures compare financial year 2022 with financial year 2025.

The study does not assess the technical performance of AI models, does not use confidential bank data, and does not evaluate any individual bank. It is conducted independently and not as a company project.

1.6 DELIVERABLES

The study delivers:

1. An analysis of how AI adoption is changing GRC activities, role structures and skill demand in UK and EU banks.
2. A mapping of UK and EU regulatory requirements (DORA, the EU AI Act, SS1/23 and SM&CR) to workforce competence and human oversight obligations.
3. A statistical assessment, on public industry data, of the hypotheses arising from the practitioner evidence.
4. A GRC role-impact map classifying activities and roles as automated, augmented, newly created or human-retained.
5. An assessment of skill gaps and reskilling strategies, including build, buy and borrow options for GRC talent.
6. An analysis of employee behavioural responses (perceived usefulness, ease of use, trust and perceived replaceability) and their implications for adoption.

7. An integrated seven-stage workforce transformation framework.
8. Evidence-based recommendations for GRC and HR leadership in UK and EU banks.

1.7 ORGANISATION OF THE REPORT

Chapter 1 introduces the background, problem, need, objectives, scope and deliverables of the study. Chapter 2 reviews the literature on technology acceptance, trust, workplace automation, RegTech and employee perceptions of AI, sets out the regulatory context and identifies the research gap. Chapter 3 describes the research methodology, including the sequential exploratory design, data sources, instruments, sampling, propositions, hypotheses and statistical tools. Chapter 4 presents the analysis and interpretation of the interview, questionnaire and secondary data, the statistical hypothesis tests, the triangulation of evidence, the GRC role-impact map and the workforce transformation framework. Chapter 5 summarises the findings, presents recommendations and conclusions, and suggests directions for future research.

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

AI in Governance, Risk and Compliance has been studied mainly as a technological and regulatory development. The workforce consequences of that development have been examined mostly in the general management literature rather than in a banking compliance setting. This chapter draws the two streams together. It reviews eight sources from management, organisational behaviour and financial regulation, grouped into four themes: technology acceptance and trust (Section 2.2), automation and the recomposition of work (Section 2.3), RegTech and the transformation of compliance (Section 2.4), and employee perceptions of AI (Section 2.5). Section 2.6 sets out the regulatory context that makes workforce capability an obligation in UK and EU banks. Section 2.7 summarises the review, Section 2.8 identifies the research gap, and Section 2.9 links the literature to the objectives and propositions of the study.

2.2 TECHNOLOGY ACCEPTANCE AND TRUST IN AI

2.2.1 Technology Acceptance Model (Davis, 1989)

Davis (1989) established the Technology Acceptance Model (TAM). Through validated measurement scales, he showed that an individual's use of a technology is determined by two beliefs: perceived usefulness, the degree to which a person believes the system will improve job performance, and perceived ease of use, the degree to which using it will be free of effort. Perceived usefulness was the stronger determinant of use, and ease of use acted largely through it: a system that is easy to use but does not help the user do the job is unlikely to be adopted.

TAM is relevant to GRC because AI tools are only valuable if compliance and risk professionals actually rely on them. A tool that is technically capable but perceived as unhelpful will be worked around, leaving the bank paying for both the technology and the manual process it was meant to replace. The present study adapts Davis's usefulness and ease-of-use constructs for the questionnaire (Constructs A and B).

2.2.2 Human trust in AI (Glikson and Woolley, 2020)

Glikson and Woolley (2020) reviewed two decades of empirical research on human trust in AI. They concluded that the successful integration of AI into organisations depends critically on workers' trust. They distinguished cognitive trust, which is shaped by the tangibility, transparency, reliability and immediacy behaviours of the technology, from emotional trust, which is associated with anthropomorphism. Their review also showed that the way AI is represented (as a robot, a virtual agent or embedded in software) affects how trust develops.

Two points from this review are central to the present study. First, transparency and reliability are separate drivers: a user may find outputs reliable but still not understand how they were produced. In GRC, where staff must be able to explain and defend a decision to a regulator or auditor, this distinction matters. Second, Glikson and Woolley observed that the evidence base relies heavily on short-term, small-sample experimental studies, and that trust is likely to develop differently in longer-term, high-stakes field settings. Regulated banking GRC is such a setting. The questionnaire's trust construct (Construct C) is adapted from this review.

2.3 AUTOMATION AND THE RECOMPOSITION OF WORK

2.3.1 Substitution and complementarity (Autor, 2015)

Autor (2015) examined the history and future of workplace automation and asked why, despite centuries of labour-saving technology, there are still so many jobs. He argued that automation substitutes for labour in some tasks while complementing it in others. Machines take over routine, codifiable tasks, but this raises the value of the

tasks that remain, particularly those requiring judgement, problem solving, adaptability and interpersonal skill. Occupations are therefore recomposed rather than simply eliminated, and the gains from automation depend on how well human and machine tasks are combined.

This distinction underpins the view that AI in GRC leads mainly to role redesign rather than to wholesale job loss. A transaction-monitoring analyst whose first-level alert triage is automated does not disappear; the role shifts towards investigating the alerts that remain and documenting decisions. Autor's framework is the basis for the first research proposition (P1): routine GRC tasks are automated first, while judgement-intensive tasks are augmented.

2.3.2 Susceptibility to computerisation (Frey and Osborne, 2017)

Frey and Osborne (2017) estimated the probability of computerisation for 702 occupations, using expert task assessments and a classification model applied to occupational task data. They concluded that about 47% of total United States employment is at high risk of computerisation over the coming decades. Susceptibility was highest for routine and codifiable tasks and was negatively related to wages and educational attainment. They identified perception and manipulation, creative intelligence and social intelligence as the main bottlenecks to computerisation.

Their findings are directly relevant to GRC, where regulatory monitoring, transaction screening, data aggregation and compliance reporting contain substantial routine content, while interpreting regulatory intent, challenging risk decisions and advising senior management depend on the bottleneck capabilities they describe. Read together with Autor (2015), the study suggests that GRC exposure should be assessed at the level of activities rather than whole job titles.

2.3.3 Technology, work and organisations (Cascio and Montealegre, 2016)

Cascio and Montealegre (2016) reviewed how technology is changing work and organisations from an organisational psychology and organisational behaviour perspective. They examined the number and types of jobs affected by technological

development and the resulting worker dislocation, illustrated the effects through technologies such as electronic monitoring systems and robots, and set out implications for six areas of human resource management. Their central argument is that technological change is a matter of organisational design and HR management, not of technology alone.

For the present study, this review justifies treating AI in GRC as a workforce management question. Decisions about job design, selection, training, performance management and employee well-being determine whether the technology produces its intended benefits. It also supports the third proposition (P3), that new technology changes the skill profile organisations need.

2.4 REGTECH AND THE TRANSFORMATION OF COMPLIANCE

2.4.1 The concept of RegTech (Arner, Barberis and Buckley, 2017)

Arner, Barberis and Buckley (2017) introduced the concept of RegTech, the use of technology for regulatory monitoring, reporting and compliance. They argued that regulatory change and technological development following the 2008 Global Financial Crisis are together transforming how compliance is performed and how supervisors work. In their view, RegTech had so far largely digitised manual compliance processes, and its real potential lies in near real-time, proportionate regulation built around data, digital identity and automated reporting. This requires a reconceptualisation of financial regulation itself.

Their work establishes that compliance work is being restructured rather than merely supported by technology. If compliance shifts from periodic, sample-based checking to continuous, data-driven monitoring, the people who perform it need different skills. Arner et al. do not, however, examine those people directly; the workforce is implicit in their analysis.

2.4.2 The European route to RegTech (Buckley, Arner, Zetzsche and Weber, 2020)

Buckley, Arner, Zetzsche and Weber (2020) analysed the European Union's route to RegTech and identified four pillars: extensive post-crisis reporting requirements, strict data protection rules, the facilitation of open banking, and a legislative framework for digital identity. They argued that European financial services and data protection reforms have unintentionally driven the adoption of regulatory technology by banks, supervisors and regulators.

This analysis is important for the present study because it shows regulation acting as a driver of technology adoption in European compliance functions, not only as a constraint on it. Regulation shapes what is automated and, by extension, what remaining human work looks like. This supports the second proposition (P2), that regulation converts competence and oversight into compliance obligations.

2.5 EMPLOYEE PERCEPTIONS OF AI AND AUTOMATION

2.5.1 STARA awareness (Brougham and Haar, 2018)

Brougham and Haar (2018) studied employees' perceptions of smart technology, artificial intelligence, robotics and algorithms (STARA). They developed a measure of the extent to which employees believe their job could be replaced by such technologies. Using a mixed-methods design with 120 employees, they found that greater STARA awareness was negatively related to organisational commitment and career satisfaction, and positively related to turnover intentions, cynicism and depression.

Their study shows that the workforce consequences of automation are behavioural and managerial, and are not confined to headcount. Employees who expect to be replaced may disengage or leave before any role is actually removed, which is costly in GRC, where experienced staff hold knowledge that is needed to validate automated outputs. The questionnaire's perceived replaceability and organisational response constructs (Constructs D and G) are adapted from this study.

2.6 REGULATORY CONTEXT: WORKFORCE CAPABILITY AS AN OBLIGATION

The literature above must be read against a regulatory environment in which workforce capability has become an obligation. Four instruments are central.

- **EU Digital Operational Resilience Act (DORA), Regulation (EU) 2022/2554.** The management body bears ultimate responsibility for ICT risk and must keep its knowledge and skills up to date (Article 5). Firms must provide ICT security awareness and digital operational resilience training for all staff, including senior management (Article 13).
- **EU Artificial Intelligence Act, Regulation (EU) 2024/1689.** Providers and deployers must take measures to ensure a sufficient level of AI literacy among staff (Article 4). High-risk AI systems, which include creditworthiness assessment, must be subject to effective human oversight by people who understand the system's limits, are aware of automation bias and can override it (Article 14). The Digital Omnibus on AI (2026) retained the literacy duty as a duty of means and deferred some high-risk obligations.
- **PRA Supervisory Statement SS1/23, Model risk management principles for banks.** Firms must designate an accountable senior manager for the model risk framework, document roles across the model lifecycle with required skills and experience, and ensure that validators have the expertise and standing to challenge. The principles apply to AI and machine-learning models.
- **UK Senior Managers and Certification Regime (SM&CR).** There is no dedicated AI senior manager, but existing senior management functions and conduct rules apply to AI use, including the duty to take reasonable steps. AI-assisted decisions therefore remain the personal responsibility of named individuals (FCA, 2024).

Together these instruments mean that banks must be able to demonstrate that their staff are competent to understand, challenge and supervise AI-driven risk and compliance outputs. Competence and human oversight are no longer discretionary HR preferences; they are conditions of regulatory compliance.

2.7 SUMMARY OF THE LITERATURE REVIEWED

Table 2.1 Summary of the literature reviewed

Author (year)	Focus	Key finding	Use in this study
Davis (1989)	Technology acceptance	Perceived usefulness and ease of use determine use; usefulness is stronger	Questionnaire Constructs A and B; H1, H2
Autor (2015)	Workplace automation	Automation substitutes routine tasks and complements judgement; jobs are recomposed	Proposition P1; role-impact map
Cascio and Montealegre (2016)	Technology and organisations	Technological change is an organisational design and HR issue	Workforce management lens; P3
Frey and Osborne (2017)	Susceptibility to computerisation	Routine, codifiable tasks are most exposed	Proposition P1; activity-level analysis
Arner, Barberis and Buckley (2017)	RegTech	Compliance work itself is being restructured by technology	Context for Objective (i)
Brougham and Haar (2018)	Employee perceptions (STARA)	Perceived replaceability lowers commitment and raises turnover intention	Questionnaire Constructs D and G; H3; Objective (iv)
Buckley, Arner, Zetzsche and Weber (2020)	EU RegTech	Regulation drives technology adoption in EU compliance	Proposition P2; Objective (ii)
Glikson and Woolley (2020)	Trust in AI	Cognitive trust rests on transparency and reliability; field evidence is scarce	Questionnaire Construct C; H4

2.8 RESEARCH GAP

The literature establishes three positions that have not yet been brought together.

1. **The RegTech literature** (Arner et al., 2017; Buckley et al., 2020) explains how compliance functions and regulation are being transformed by technology, but treats the workforce only implicitly.

2. **The work and employment literature** (Autor, 2015; Cascio & Montealegre, 2016; Frey & Osborne, 2017; Brougham & Haar, 2018) establishes that automation recomposes rather than removes jobs, that routine tasks are most exposed, and that perceived replaceability carries measurable costs in commitment and turnover. It is, however, cross-industry rather than specific to banking GRC.

3. **The technology acceptance and trust literature** (Davis, 1989; Glikson & Woolley, 2020) identifies the individual-level determinants of adoption but, as Glikson and Woolley note, rests largely on experimental settings rather than high-stakes regulated environments.

The gap addressed by this study lies at the intersection of the three: how AI-driven transformation of GRC in UK and EU banks is reshaping workforce management decisions on role design, skills and staffing, and how employees in a regulated, accountability-bound function adapt to it. No reviewed source combines these perspectives, and none considers how DORA, the AI Act, SS1/23 and SM&CR convert workforce capability into a regulatory requirement.

2.9 LINKING THE LITERATURE TO THE STUDY

The review leads directly to the objectives and propositions of the study, as shown in Table 2.2.

Table 2.2 Link between literature, objectives and propositions

Literature	Objective	Proposition / hypothesis
Autor (2015); Frey and Osborne (2017)	(i) Activities automated or augmented; role demand	P1: routine tasks are automated first; judgement work is augmented
Arner et al. (2017); Buckley et al. (2020); regulatory texts	(ii) Regulation, oversight and competence	P2: regulation converts competence and oversight into compliance obligations
Cascio and Montealegre (2016); Autor (2015)	(iii) Skill gaps and reskilling	P3: AI creates demand for a hybrid technical and regulatory skill profile
Autor (2015), extended by interview evidence	(iii) and (v)	P4: loss of institutional knowledge is a material risk of restructuring
Davis (1989); Glikson and Woolley (2020); Brougham and Haar (2018)	(iv) Employee behavioural responses	H1–H4 on usefulness, ease of use, role change and conditional trust

The next chapter explains how these propositions and hypotheses were examined.

CHAPTER 3

METHODOLOGY

3.1 TYPE OF PROJECT

The project is an exploratory, independent research study. It is not a company project, and no internal bank documentation or workforce data were used. The study follows a **sequential exploratory mixed-methods design**, in which qualitative evidence is collected first and is then used to frame quantitative tests on a wider population. The design runs in three steps:

1. **Interviews and questionnaire set the hypotheses.** Written semi-structured interviews with GRC practitioners and senior professionals produced ten themes (T1–T10). A structured questionnaire then measured how strongly practitioners agree with those themes, reported as descriptive means and standard deviations.
2. **Public data test the hypotheses.** Each testable theme was translated into a hypothesis (H5–H10), which was tested statistically on independent public industry data: Eurostat enterprise statistics, bank annual reports, job adverts and published industry surveys (datasets D1–D4).
3. **Triangulation.** Practitioner views, public-data results, regulatory analysis and the reviewed literature are compared side by side to assess four research propositions (P1–P4) and to build the workforce transformation framework.

The statistical tests therefore answer one question: do the patterns that practitioners describe also hold across the wider industry? The interviews and questionnaire decided which tests were run. The public datasets are independent sources

and were not produced from the primary data. This design suits the research problem, which is exploratory, and the practical constraint that banks do not publicly disclose workforce practices at the level of detail required.

3.2 DATA REQUIREMENT

The study requires four categories of evidence, shown in Table 3.1.

Table 3.1 Data requirement, sources and objectives served

Data category	Source	Objective served
1. AI adoption in GRC: activities automated or augmented	Interviews; supervisory surveys; Eurostat (D1); bank annual reports (D2); published surveys (D4)	(i)
2. Role and skill change: job content, new roles, skill demand	Interviews; public job adverts (D3); labour-market and professional-body reports	(i), (iii)
3. Regulatory requirements on workforce competence and human oversight	UK and EU regulatory and supervisory texts; bank annual reports (D2)	(ii)
4. Employee and managerial perceptions: usefulness, ease of use, trust, replaceability, skills, oversight, organisational response	Interviews and questionnaire (primary)	(iii), (iv)

Categories 1 to 3 are obtained mainly from secondary sources, while category 4 requires primary data. All secondary sources are publicly available, and no confidential, client or employer-specific information is used.

3.3 TARGET RESPONDENTS

The population is professionals working in or with GRC functions in banks and banking service providers, including global capability centres, operating in the UK and the EU. This covers risk, compliance, internal audit, financial crime, GRC technology, and HR business partners supporting these functions.

The respondents reached span internal audit, risk, GRC technology, operational resilience, enterprise and data architecture supporting GRC service lines, and senior consulting leadership. They include analyst, manager and senior manager levels, and their responsibilities cover the UK, the EU or both. Reported AI maturity in

respondents' areas ranged from no AI use (with established automation), through pilots only, to AI used in some processes. No respondent reported extensive use.

3.4 SAMPLING METHODS

3.4.1 Primary data

Purposive sampling, supplemented by snowball referral, was used. Purposive sampling is appropriate because the study requires respondents with direct exposure to AI-enabled GRC processes. A random sample of banking employees would largely comprise people unable to answer the questions. The selection criteria were:

i. experience in a GRC or GRC-supporting role;

ii. current or recent involvement with UK or EU banking; and

iii. direct or supervisory exposure to AI-enabled or automated GRC processes.

Interviews were continued until no materially new themes emerged on the core questions, which indicates thematic saturation within a homogeneous senior professional group. The questionnaire serves as a validation of the interview themes and not as a basis for statistical generalisation. The primary data therefore support analytical rather than statistical generalisation.

3.4.2 Secondary data used for statistical testing

Four public datasets were assembled to test the hypotheses. Their coverage is shown in Table 3.2.

Table 3.2 Public datasets used for hypothesis testing

Dataset	Source	Unit and coverage	Role in the study
D1	Eurostat, enterprise AI use (isoc_eb_ai) and ICT specialists in employment (isoc_sks_itspt)	27 EU countries; 2021, 2023, 2024 and 2025	Economy-wide AI adoption and ICT-skills context

Dataset	Source	Unit and coverage	Role in the study
D2	Annual reports from banks' official investor-relations websites	Frame of 29 large UK and EU banks (7 UK, 22 EU); FY2022 and FY2025. 42 of 58 report-years coded; 19 banks with both years	Change in AI, reskilling, AI-governance and regulation disclosure
D3	Reed.co.uk job adverts	163 UK financial-services GRC adverts, of which 83 state an annual salary	Skills and pay in AI-exposed roles
D4	Published surveys: Bank of England and FCA (2022, 2024), IIA Vision 2035, Thomson Reuters Future of Professionals 2025	Published proportions with their sample sizes	Industry adoption and skills-constraint evidence

D1 is a census of EU member states, so no sampling was needed. The Eurostat survey excludes the financial sector, so D1 is used as economy-wide context rather than as direct banking evidence.

D2 uses a purposive frame of 29 of the largest UK and EU banks, fixed before data collection. For each bank, the full English annual report (or universal registration document) for FY2022 and FY2025 was taken from the bank's own investor-relations website. Every file was checked against its title page to confirm the bank and fiscal year, and mismatched files were rejected. Report-years for which no full English report was available, or whose website blocked automated access, were recorded with a reason and excluded. This produced 42 coded report-years and 19 banks with both years, which form the paired sample. Banks are shown by code in the analysis tables, and the banks included are listed in Appendix 3.

D3 was collected from the public Reed.co.uk job board using fixed GRC search terms (for example, AI risk, compliance and financial crime roles), taking the first results pages for each term, in October 2026. Only financial-services adverts were kept, and employer names were not recorded.

D4 consists of published proportions and sample sizes from public industry and supervisory surveys. No new survey data were collected.

3.5 RESEARCH INSTRUMENTS

3.5.1 Semi-structured interview guide

The interview guide contains eleven open-ended questions in five sections: (1) current use of AI in GRC work; (2) changes in role content and team structure; (3) regulation, human oversight and accountability; (4) skills and reskilling; and (5) employee response and the management of change, followed by a closing open question. Because of respondents' seniority and availability, the guide was administered as a written interview through an online form. This allowed considered, detailed answers. The form is reproduced in Appendix 1.

3.5.2 Structured questionnaire

The questionnaire has a profile section (function, level, experience, jurisdiction, type of organisation and extent of AI use), followed by 25 items rated on a five-point Likert scale (1 = Strongly disagree to 5 = Strongly agree) and two optional open questions. The items are grouped into seven constructs, shown in Table 3.3. The full questionnaire is reproduced in Appendix 2.

Table 3.3 Questionnaire constructs

Construct	Items	Adapted from
A. Perceived usefulness	7–10	Davis (1989)
B. Perceived ease of use	11–13	Davis (1989)
C. Trust in AI output	14–18	Glikson and Woolley (2020)
D. Perceived replaceability	19–21	Brougham and Haar (2018)
E. Skills and reskilling	22–25	Study-specific
F. Human oversight and accountability	26–27	Study-specific, grounded in DORA, the AI Act, SS1/23 and SM&CR
G. Organisational response and outcomes	28–31	Brougham and Haar (2018)

3.5.3 Method of preparing the instruments

The interview guide was derived from the secondary objectives and the gaps identified in the literature review. The questionnaire was then built from the interview themes, using constructs adapted from the reviewed literature rather than newly invented scales, so that findings can be compared with published work. Each instrument

item is mapped to an objective, as shown in Table 3.4. Both instruments include an informed-consent statement.

Table 3.4 Mapping of instruments to objectives

Objective	Interview questions	Questionnaire items
(i) GRC activities automated or augmented; role demand	1–4	6, 20, 32
(ii) Regulatory requirements, oversight and competence	5–6	16, 26, 27
(iii) Skill gaps and reskilling strategies	7–8	22–25, 33
(iv) Employee behavioural responses	9–10	7–19, 21, 28–31
(v) Framework and recommendations	Synthesis of all	Synthesis of all

3.6 RESEARCH PROPOSITIONS AND HYPOTHESES

3.6.1 Research propositions

Because the design is exploratory and qualitatively dominant, the study is guided by four research propositions, assessed through triangulation (Table 3.5).

Table 3.5 Research propositions

No.	Proposition	Main evidence
P1	Routine, rules-based GRC tasks are automated first, while judgement-intensive tasks are augmented rather than replaced.	Interviews; job-description analysis; H5, H8, H10a
P2	Regulation converts workforce competence and human oversight into compliance obligations.	Regulatory texts; interviews; H10a (regulation), H10b, H10d
P3	AI adoption in GRC creates demand for a hybrid skill profile combining technical and data fluency with domain and regulatory judgement.	Interviews; Construct E; H5b, H6, H7, H9
P4	Loss of institutional knowledge is a material risk of AI-led workforce restructuring in GRC.	Interviews (emergent theme); secondary evidence

3.6.2 Questionnaire hypotheses (descriptive)

Four directional hypotheses are assessed on the questionnaire by comparing the construct or item mean with the neutral scale midpoint of 3:

- **H1:** GRC professionals perceive AI-enabled tools as useful (Construct A mean > 3).
- **H2:** GRC professionals perceive AI-enabled tools as easy to use (Construct B mean > 3).
- **H3:** GRC professionals expect their roles to change substantially because of AI (item 20 mean > 3).
- **H4:** Trust in AI output is conditional: comfort in relying on output without re-performing the work (item 17) is rated lower than the reliability of the output (item 15).

The questionnaire was administered to a small, purposive group of senior practitioners as a validation instrument. With a sample of this kind, significance tests and reliability coefficients such as Cronbach's alpha would not be meaningful. H1–H4 are therefore assessed descriptively, and the statistical testing in this study is performed on the public datasets.

3.6.3 From interview themes to public-data hypotheses

Table 3.6 shows how each interview theme, together with its questionnaire signal, was translated into a hypothesis that could be tested on public data.

Table 3.6 Translation of interview themes into public-data hypotheses

Interview theme	What practitioners said	Questionnaire signal (mean, 1–5)	Public-data hypothesis
T1. Automation precedes AI; pilot-to-production gap	AI is spreading but is often still at pilot stage	Role will change (item 20): 4.33	H5a, H8: AI adoption is rising
T2/T3. Routine work absorbed; roles recomposed	Rules-based tasks are automated first; roles shift towards review	Usefulness (Construct A): 3.58	H10a (AI density); H6: AI-exposed roles have a different skill mix
T4/T5. Expert review and accountability	Humans must be able to challenge AI; senior accountability matters	Can defend AI decision (item 16): 4.33; accountable people knowledgeable (item 26): 4.00	H10a (AI-governance density); H10b: named AI governance bodies
T6. Regulation felt unevenly	DORA, AI Act and SS1/23 matter, but unevenly	Oversight requirements clear (item 27): 3.67	H10a (regulation density); H10d: UK vs EU

Interview theme	What practitioners said	Questionnaire signal (mean, 1–5)	Public-data hypothesis
T7. Hybrid skill profile	Data, AI, cyber and automation skills rising alongside judgement	Data skills (item 24): 4.00; judgement (item 25): 4.00	H5b, H6, H7: skills and pay in AI roles
T8/T9. Knowledge loss; build, buy, borrow	Training lags; specialist gaps persist	Training adequate (item 23): 3.33	H9: skills constraint; H10a (reskilling); H10c: reskilling vs AI intensity
T10. Anxiety and weak change management	Concern about jobs; weak communication	Replaceability concern (Construct D): 3.89; organisational response (Construct G): 3.58	Not testable in public data; questionnaire only

3.6.4 Statistical hypotheses

The statistical hypotheses tested on public data are listed in Table 3.7.

Table 3.7 Statistical hypotheses tested on public data

No.	Null hypothesis (H0)	Alternative hypothesis	Data
H5a	AI adoption did not increase between 2021 and 2025	AI adoption increased	D1
H5b	ICT-specialist share is not associated with AI adoption	Positive association	D1
H6	Skill demand is the same in AI-exposed and other adverts	Skill demand differs	D3
H7	AI-related adverts do not offer higher pay	AI-related adverts offer higher pay	D3
H8	The share of UK financial firms using AI did not rise between 2022 and 2024	The share rose	D4
H9	At most half of firms or professionals see skills or talent as a constraint	More than half do	D4
H10a	AI, reskilling, AI-governance and regulation term density in bank reports did not rise from FY2022 to FY2025	Density rose	D2
H10b	The share of banks naming an AI governance body did not change	The share changed	D2
H10c	FY2025 AI density is not correlated with reskilling density	They are correlated	D2
H10d	UK and EU banks do not differ in FY2025 densities	They differ	D2

3.7 DATA PROCESSING

Interviews. Written responses were read in full and open-coded line by line. Codes were grouped into themes aligned to the secondary objectives, and illustrative verbatim quotations were selected for each theme. Coding was performed manually in Microsoft Excel.

Questionnaire. Responses were exported from the online form. Item and construct means and standard deviations were calculated. Construct D is reported in the direction of concern (a higher score means more concern). Item 30 (considering leaving) is reverse-coded within Construct G so that a higher score means a more positive organisational response. Responses were screened for uniform answering using the reverse-worded items, and results are interpreted with that caution in mind.

D1. Country-level values were downloaded from the Eurostat API for enterprises with 10 or more employees and with 250 or more employees.

D2. Each annual report was converted from PDF to text, and fixed, case-insensitive regular expressions were run over the full text to count four groups of terms: AI, reskilling, AI governance and regulation (the AI Act, DORA and SS1/23). Each count was divided by the report's total word count and multiplied by 10,000, giving a density per 10,000 words, so that long and short reports can be compared fairly. A binary variable records whether the report names a dedicated AI governance body or role. The coding dictionary is given in Appendix 3, and the variables are defined in Table 3.8.

D3. Advert text was flagged for mentions of AI or machine learning ("AI-exposed") and for six skill groups: data, regulatory, cyber, automation, certification and governance. For pay analysis, the midpoint of the advertised annual salary was used. Daily and hourly rates were excluded.

D4. Published proportions and their sample sizes were recorded with the page reference of the source statement. Items without a published sample size were reported descriptively and not tested.

Table 3.8 Variables used in the statistical analysis

Variable	Definition
AI adoption, 10+ / 250+ (D1)	% of enterprises with 10+ / 250+ employees using at least one AI technology
ICT specialists (D1)	ICT specialists as % of total employment
AI density (D2)	Mentions of AI terms (artificial intelligence, machine learning, generative AI, large language models, "AI") per 10,000 words
Reskilling density (D2)	Mentions of reskilling, upskilling, retraining, digital skills, AI or data literacy, skills gap or development per 10,000 words
AI-governance density (D2)	Mentions of AI governance, responsible or ethical AI, human oversight, human-in-the-loop, explainability and model risk per 10,000 words
Regulation density (D2)	Mentions of the EU AI Act, DORA or digital operational resilience, and PRA SS1/23 per 10,000 words
AI governance body (D2)	1 if the report names a dedicated AI governance body or role (e.g. chief data or AI officer, AI council or committee, responsible-AI office), else 0
AI-exposed (D3)	1 if the advert mentions AI or machine learning
Skill flags (D3)	1 if the advert asks for data, regulatory, cyber, automation, certification or governance skills
Annual salary (D3)	Midpoint of the advertised annual salary (£)

3.8 TOOLS FOR ANALYSIS

The qualitative and descriptive analysis used Microsoft Excel. The statistical analysis used Python (pandas, SciPy, statsmodels and matplotlib), and all results can be reproduced from the project's analysis script. The significance level is $\alpha = 0.05$. Tests are one-sided where the hypothesis predicts a direction. The choice of test for each hypothesis is explained below.

- **Thematic analysis** (interviews): open coding grouped into themes by objective.
- **Content analysis** (regulatory texts, supervisory publications and job descriptions): activities coded as automated, augmented or unaffected, and skills coded as technical, analytical, regulatory or behavioural.
- **Descriptive statistics** (questionnaire): item and construct means and standard deviations compared with the midpoint of 3.
- **Paired t-test with Wilcoxon signed-rank test** (H5a, H10a): the same countries or banks are measured twice, so the comparison is paired. The t-test compares mean change, and the Wilcoxon test confirms the result

without assuming normality, which matters for small, skewed samples such as D2 densities.

- **Friedman test** (H5a): a non-parametric repeated-measures test across all four Eurostat years.
- **Pearson and Spearman correlation and simple OLS regression** (H5b): these measure the strength of the link between ICT skills and AI adoption. Spearman is the rank-based check.
- **Chi-square or Fisher exact test with Holm correction** (H6): skill flags are yes/no variables. Fisher's test is used when expected cell counts are small, and the Holm correction controls for testing six skills at once.
- **Mann–Whitney U test with Welch t-test** (H7, H10d): these compare two independent groups. Salaries and densities are skewed and the groups are small, so the rank test is the main test. The Holm correction is applied across the four D2 densities.
- **Two-proportion and one-sample z-tests** (H8, H9): only published proportions and sample sizes are available.
- **Exact McNemar test** (H10b): a paired yes/no outcome for the same banks in two years.
- **Spearman rank correlation** (H10c): a rank correlation that is robust to outliers in a small sample.
- **Triangulation**: the evidence streams are compared side by side for each proposition. The questionnaire and the public data measure different units (individuals, as against countries, banks and adverts), so they are compared, not pooled into one test.

3.9 ASSUMPTIONS, CONSTRAINTS AND LIMITATIONS

Assumptions. Respondents answered honestly and from their professional experience. The language of bank annual reports reflects, at least in part, the priorities of bank management. Job adverts reflect the skills employers are currently seeking.

Constraints. The study was conducted without institutional access, so no internal bank data were available. Senior respondents had limited time, which is why the interviews were collected in written form. Some bank websites block automated download, which limited the number of paired bank reports.

Limitations.

1. **Primary sample.** The interviews and questionnaire come from a small, purposive group with a technology-leaning profile. They support analytical, not statistical, generalisation, and the questionnaire results are descriptive only.
2. **Self-reported perceptions.** Trust, usefulness and replaceability are self-reported and may be affected by social desirability.
3. **D1** covers the whole economy and excludes the financial sector, so it is context only.
4. **D2** counts words, not actions. A mention of AI or reskilling does not show what a bank actually does, and results depend on report length, language and PDF text extraction. Only 19 of the 29 banks had both years available, and UK banks are few (six in FY2025), so the regional tests have low power. One bank's financial year ends in April and is mapped to the nearest year.
5. **D3** comes from one UK job board on a single collection date (a convenience sample). It contains no EU adverts and few AI-titled adverts, so the pay result is indicative only.
6. **D4** is secondary published data, and survey waves use different samples.
7. **Multiple testing.** Holm corrections are applied within families of tests (H6, H10d), but other p-values are unadjusted.
8. **Cross-sectional design and fast-moving context.** Primary data capture one point in an ongoing transition. The AI Act is being phased in and has been amended, so specific requirements may change.

9. **Jurisdictional breadth.** The UK and EU are treated as one combined scope, so national differences within the EU are not examined.

3.10 ETHICAL CONSIDERATIONS

Participation was voluntary, and every respondent gave informed consent to the use of anonymised responses. No respondent, employer or client is identified in this report. Respondents were asked to speak to general practice and not to share confidential or proprietary information. Primary data are stored securely and used only for academic purposes. All secondary data are publicly available. Banks are named only in Appendix 3, in connection with their published annual reports.

CHAPTER 4

DATA ANALYSIS AND INTERPRETATION

4.1 INTRODUCTION

This chapter presents the analysis in the order of the research design. Section 4.2 describes the respondents. Section 4.3 presents the thematic analysis of the interviews, and Section 4.4 the descriptive analysis of the questionnaire. Section 4.5 analyses the secondary evidence on AI adoption, regulation, job descriptions and workforce practices. Section 4.6 reports the statistical tests of hypotheses H5–H10 on public data. Section 4.7 triangulates all the evidence against the propositions. Sections 4.8 and 4.9 present the two main outputs of the study: the GRC role-impact map and the workforce transformation framework. Section 4.10 maps the deliverables to the analysis.

4.2 PROFILE OF RESPONDENTS

Table 4.1 summarises the profile of the primary-data respondents. In line with the anonymity commitments given to participants, the profile is described qualitatively.

Table 4.1 Profile of respondents

Element	Interviews	Questionnaire
Functions	Enterprise data architecture and enterprise architecture supporting GRC service lines; technology professionals working on operational resilience; senior consulting leadership	Internal audit, risk and GRC technology
Levels	Senior professionals and leaders	Analyst, manager and senior manager
Jurisdiction	UK and dual UK/EU responsibilities	UK, EU and dual jurisdictions
Organisation type	Banks, banking service providers and consultancies serving banks	Banking service providers and global capability centres

Element	Interviews	Questionnaire
AI maturity in own area	From no AI use with established automation, through pilots only, to AI used in some processes; no respondent reported extensive use	
Collection	Written interview through an online form, September 2026	Online questionnaire, September 2026

The respondent group is senior and leans towards technology and architecture roles. This gives an informed view of how AI is being introduced, but under-represents first-line compliance staff and HR business partners. The findings are interpreted with this in mind.

4.3 THEMATIC ANALYSIS OF INTERVIEW DATA

Open coding of the written interviews produced ten themes. They are presented below by objective, with illustrative verbatim evidence.

4.3.1 Objective (i): activities automated or augmented, and role demand

Table 4.2 Interview themes for Objective (i)

Theme	Finding	Illustrative evidence (verbatim)
T1. Automation precedes AI; a wide pilot-to-production gap	Rules-based automation of regulatory monitoring and reporting, governance workflows, control-testing support and data aggregation is established. AI (document analysis, regulatory change assessment, enhanced risk scoring, compliance monitoring) is largely at pilot stage. Production AI uses are specific: reconciliation and documentation of complex process flows, and identifying vulnerabilities and developing test scenarios for operational resilience.	"I don't actually see it working in my experience. I have only ever seen pilot."

Theme	Finding	Illustrative evidence (verbatim)
T2. Routine work is absorbed; judgement work resists	Most affected: data collection and consolidation for compliance reporting, control monitoring, regulatory reporting workflows, MI and dashboards, data mapping and design diagrams. Resistant: challenging risk decisions, control design, agreeing remediation, assurance to senior stakeholders and regulators, and operations-specific knowledge of complex flows and regulation.	"Tasks with structured inputs, defined rules and repeatable processes were generally easier to automate. Activities involving ambiguity, interpretation of regulatory intent, business context, or balancing competing risks continued to rely heavily on experienced practitioners."
T3. Roles are recomposed; testing roles contract first	Material time savings in data analysis, source-file comparison, cross-format reconciliation, requirements interviewing, and design and architecture diagramming. Freed capacity is redirected to review and higher-order design.	"GRC Testing Roles have disappeared."

Interpretation. The pattern matches Autor's (2015) distinction between substitution and complementarity. Tasks with structured inputs and fixed rules are substituted, while tasks that require interpretation and judgement are complemented. The gap between pilots and production also means that much of the workforce impact is still ahead.

4.3.2 Objective (ii): regulation, human oversight and competence

Table 4.3 Interview themes for Objective (ii)

Theme	Finding	Illustrative evidence (verbatim)
T4. Expert human review is non-negotiable	AI output is not reliably accurate word for word, depends on input data quality and can hallucinate. Meaningful challenge needs business and regulatory knowledge, understanding of the data, awareness of AI failure modes and the confidence to apply judgement.	"AI output needs human and expert review to be used." / "Inaccurate insights and outputs can result in reputational damage and financial penalties."
T5. An accountability gap at the top	Senior executives accountable under SM&CR and DORA may lack the understanding needed to challenge AI, creating a risk of automation bias.	"C-level execs really do not understand what the AI is, and thus are likely to simply trust it."

Theme	Finding	Illustrative evidence (verbatim)
T6. Regulation is felt unevenly in practice	Some practitioners see no practical regulatory effect yet, and one judged the regulatory environment inadequate to protect. Others see regulatory exposure as the factor that sustains human validation.	"At the moment, the regulatory environment is woefully lacking in the ability to protect." / "None so far."

Interpretation. Practitioners accept that human review is required, but the people who are formally accountable may be the least able to provide it. This is the risk that Glikson and Woolley (2020) describe: where users cannot see into a system, they tend either to distrust it entirely or to trust it uncritically.

4.3.3 Objective (iii): skill gaps and reskilling

Table 4.4 Interview themes for Objective (iii)

Theme	Finding	Illustrative evidence (verbatim)
T7. A hybrid skill profile	AI, data, cyber, automation and resilience skills are increasingly critical. The largest gaps are in AI risk, analytics and risk quantification. Reviewer skills (attention to detail, ability to parse large volumes of generated content) are rising in importance.	"AI, data, cyber, automation, and resilience skills are increasingly critical, with gaps in AI risk, analytics, and risk quantification."
T8. Loss of institutional knowledge is the hidden risk	Critical knowledge sits with long-serving subject-matter experts. Cutting headcount before capturing it undermines the capacity to validate AI.	"Too much business knowledge exists in people's heads, and if that knowledge is lost, it becomes much harder to validate, challenge and govern AI-generated outputs effectively. I would prioritise knowledge capture and transfer before any significant workforce reductions."
T9. Build, buy and borrow, with specialist gaps persisting	Training existing staff, targeted hiring and vendor or technology support are combined. Training works for general tool use, but specialist gaps remain. Suggested remedies: dedicated learning modules for each regulation in each domain, and prompts pre-embedded in AI tools.	"A mix of training, targeted hiring, and technology/vendor support is helping, but specialist skill gaps remain."

Interpretation. The skills required are hybrid: technical fluency is needed to understand AI output, and domain judgement is needed to challenge it. Theme T8 is an emergent finding that was not anticipated in the literature review. It means that the experts whose routine work is automated are also the people needed to validate the automation.

4.3.4 Objective (iv): employee behavioural responses

Table 4.5 Interview themes for Objective (iv)

Theme	Finding	Illustrative evidence (verbatim)
T10. Ease of use welcomed; anxiety and weak change management persist	Confidence is growing alongside concern about job impact and accuracy. Junior staff are particularly worried. Trust depends on the ability to understand and validate output. Business change management is described as weak. Where management did well, it communicated and demonstrated how AI helps daily work.	"People have embraced the ease of use and accessibility." / "There have been many low level resources very worried." / "My experience is that Business Change Management is Weak."

Interpretation. The response to AI is mixed rather than hostile. Ease of use supports adoption, as TAM predicts (Davis, 1989), but anxiety among junior staff is consistent with the STARA effects described by Brougham and Haar (2018), and weak change management amplifies it.

4.4 ANALYSIS OF QUESTIONNAIRE DATA (DESCRIPTIVE STATISTICS)

The questionnaire results are reported as means and standard deviations on the five-point scale, where 3 is the neutral midpoint. They are descriptive results from a small purposive sample and are not generalised.

Table 4.6 Item-level descriptive statistics (scale 1–5)

Construct	Item	Mean	SD
A. Usefulness	7. AI improves my performance	4.00	1.00
	8. AI increases the volume of work I can complete	3.33	0.58

Construct	Item	Mean	SD
	9. AI improves the quality and consistency of outputs	3.33	1.15
	10. Overall, AI tools are useful in my job	3.67	0.58
B. Ease of use	11. Tools are easy to use	4.33	0.58
	12. Easy to understand what tools are doing	3.33	0.58
	13. Learning did not require excessive effort	4.33	0.58
C. Trust	14. I understand how tools arrive at outputs	3.67	0.58
	15. Outputs are reliable	3.67	0.58
	16. I could defend an AI-assisted decision to a regulator or auditor	4.33	0.58
	17. Comfortable relying on output without re-performing work	3.67	1.15
	18. Effective way to challenge or override AI output	4.00	1.00
D. Replaceability (higher = more concern)	19. Concerned a significant part of my role could be done by AI	3.33	1.53
	20. My role will change substantially in three years	4.33	0.58
	21. Concerned about long-term career prospects	4.00	1.00
E. Skills	22. My current skills are sufficient	3.67	1.15
	23. Organisation has provided adequate training	3.33	1.53
	24. Data and analytical skills are more important	4.00	1.00
	25. Judgement and regulatory reasoning are more important than routine processing	4.00	1.00
F. Oversight	26. Accountable people have the knowledge to supervise AI decisions	4.00	1.00
	27. Human-oversight requirements are clearly reflected in practice	3.67	1.15
G. Organisational response	28. Management communicated clearly on role impact	4.00	1.00
	29. I feel positive about AI's effect on my work	4.00	1.00
	30. I would consider leaving because of AI (higher = more intent)	3.67	1.15
	31. I feel committed to my organisation	4.00	1.00

Table 4.7 Construct composite scores

Construct	Mean	SD	Interpretation
A. Perceived usefulness	3.58	0.72	Moderately positive; strongest on individual performance, weakest on volume and quality
B. Perceived ease of use	4.00	0.33	Highest construct; tools are easy to operate and learn
C. Trust in AI output	3.87	0.50	Positive but conditional
D. Perceived replaceability (concern)	3.89	0.96	Change strongly expected; personal concern more varied
E. Skills and reskilling	3.75	1.09	Data and judgement skills rising; training adequacy is the lowest item

Construct	Mean	SD	Interpretation
F. Oversight and accountability	3.83	1.04	Oversight present; operational clarity weaker
G. Organisational response (item 30 reversed)	3.58	0.52	Communication and commitment positive; some intent to leave

Table 4.8 Assessment of questionnaire hypotheses H1–H4

Hypothesis	Measure	Mean (SD)	Result
H1: tools are useful	Construct A	3.58 (0.72)	Above the midpoint; supported descriptively
H2: tools are easy to use	Construct B	4.00 (0.33)	Above the midpoint; highest construct; supported descriptively
H3: roles will change substantially	Item 20	4.33 (0.58)	Above the midpoint, with the lowest dispersion; supported descriptively
H4: trust is conditional (item 17 < item 15)	Item 17 vs item 15	3.67 (1.15) vs 3.67 (0.58)	Equal means, so not supported on the means. Reliance without re-performance is more dispersed, and confidence in defending a decision (item 16: 4.33) exceeds comfort in relying on output (item 17: 3.67). Partly supported, with qualitative support from T4.

Key inferences from the questionnaire.

1. **Usability–transparency gap.** Ease of use (items 11 and 13: 4.33) is clearly higher than understanding what the tool is doing (item 12: 3.33). Tools are easy to operate but not easy to see into. This is consistent with interview theme T10 and with the transparency dimension of cognitive trust (Glikson & Woolley, 2020).

2. **Conditional trust.** Practitioners keep a human check in place even when they rate outputs as reliable. Comfort in relying on AI output without re-performing the work is the more dispersed rating, which supports T4.

3. **Change is accepted as inevitable.** The expectation that roles will change substantially (item 20: 4.33) is the strongest and most consistent replaceability item. Concern that one's own role could be performed by AI (item 19: SD 1.53) is the most dispersed item in the instrument. This is consistent with T10, where anxiety is concentrated among junior staff.

4. **Training lags skill demand.** The rising importance of data skills and of judgement (items 24–25: 4.00) is not matched by the adequacy of training (item 23: 3.33, SD 1.53). This supports T7 and T9.

5. **Oversight exists but is not yet operational.** The knowledge of accountable persons (item 26: 4.00) is rated higher than the clarity with which oversight requirements are reflected in practice (item 27: 3.67). This supports T5 and T6.

6. **Usefulness depends on maturity.** Usefulness was rated lower where AI was only piloted and higher where it was used in processes. This is consistent with T1 and with the Technology Acceptance Model (Davis, 1989).

In the open questions, practitioners identified human review and oversight as the part of GRC work that should not be automated. They named dedicated learning modules for each regulation within each domain as the single change that would most help staff adapt. Some answers showed signs of uniform responding across logically opposing items (for example, items 30 and 31), so these results are interpreted with caution.

4.5 ANALYSIS OF SECONDARY DATA

4.5.1 Extent of AI adoption and the skills constraint (Objective i)

Table 4.9 AI adoption and skills indicators in UK financial services

Indicator	Finding	Source
AI adoption	75% of UK financial services firms already use AI; a further 10% plan to within three years	Bank of England & FCA (2024)
Third-party dependence	One third of AI use cases are third-party implementations (up from 17% in 2022); risk and compliance, at 64%, is among the highest areas	Bank of England & FCA (2024)
Concentration	The top three third-party providers account for 73% of cloud, 44% of model and 33% of data provision	Bank of England & FCA (2024)
Understanding	46% of firms report only a "partial understanding" of the AI they use, against 34% reporting "complete understanding"	Bank of England & FCA (2024)

Indicator	Finding	Source
Skills constraint	Insufficient talent or access to skills is a large constraint for 25% of firms and a medium constraint for 32%	Bank of England & FCA (2024)
Expected growth in compliance use	A further 32% of firms expect to use AI for regulatory compliance and reporting within three years	Bank of England & FCA (2024)
Accountability	84% of firms have an accountable person for their AI framework, but most report three or more accountable persons or bodies	Bank of England & FCA (2024)

Inference. AI in GRC is spreading quickly, but largely through third-party models that firms only partly understand, with accountability split across several owners. This directly matches interview themes T1, T5 and T9.

4.5.2 Regulatory requirements mapped to workforce competence and oversight (Objective ii)

Table 4.10 Regulatory requirements and their workforce implications

Instrument	Workforce-relevant requirement	Implication for GRC workforce management
EU DORA (Reg. 2022/2554)	The management body bears ultimate responsibility for ICT risk and must maintain its knowledge and skills through regular training (Art. 5); ICT risk sits in an independent control function, with ICT auditors of sufficient expertise (Art. 6); ICT security and resilience training is compulsory for all staff and senior management (Art. 13)	Board and staff capability becomes auditable; training records and role-specific curricula become compliance evidence
EU AI Act (Reg. 2024/1689), as amended by the Digital Omnibus on AI (2026)	AI-literacy duty retained but recast as a duty of means calibrated to role and context (Art. 4); effective human oversight of high-risk systems, with overseers who understand limits, guard against automation bias and can override (Art. 14); high-risk obligations, including creditworthiness assessment, deferred	Role-based AI-literacy programmes; designated, trained overseers; the deferral is a window to build oversight capability

Instrument	Workforce-relevant requirement	Implication for GRC workforce management
PRA SS1/23	Accountable SMF for the model risk framework, including effective challenge and independent validation; roles across the model lifecycle documented with required skills and experience; validators need technical expertise, business familiarity and standing to escalate; applies to AI/ML models	AI-capable model validation and model risk roles; skills written into role definitions
UK SM&CR (with the FCA/PRA AI Update)	No dedicated AI senior manager; existing SMFs and conduct rules apply to AI use, including the duty to take reasonable steps	AI accountability falls on existing named executives, so the understanding gap (T5) becomes a personal regulatory exposure
FCA Consumer Duty	Firms must evidence good customer outcomes, including from automated decisions	Conduct and compliance staff need outcome-analytics skills
EU AML package (AMLR and AMLA, 2024)	A compliance manager at management-body level and a separate AML compliance officer with defined duties; single rulebook and direct EU supervision of selected entities	Financial crime leadership roles are redefined and split; senior AML roles must own AI-enabled monitoring outcomes
ECB supervisory priorities 2026–28	Operational resilience and ICT capabilities a standing priority; focus on banks' digital and AI strategies, governance and risk management	Supervisors will test whether AI governance is matched by people and skills

Inference (supports P2). Regulation does not prescribe headcount, but it does prescribe who must understand what, who is accountable, and that capability must be evidenced. Where AI automates a task, regulation requires a capable human to remain answerable for it.

4.5.3 Changing job descriptions and new roles (Objectives i and iii)

Table 4.11 Changing emphasis in GRC job descriptions

Role family	Traditional emphasis	Emerging emphasis in current postings
KYC / CDD analyst	Document collection, data entry, periodic refresh	Exception handling, reviewing AI-prepared case summaries, judging conflicting data, quality-assuring automated refresh

Role family	Traditional emphasis	Emerging emphasis in current postings
Transaction monitoring / sanctions analyst	High-volume manual alert disposition	Investigating AI-prioritised alerts, typology knowledge, threshold-tuning input, documenting override rationale
Compliance testing / control assurance	Sample-based manual testing, evidence chasing	Continuous control monitoring design, full-population analytics, validating automated tests; pure execution roles contracting
Regulatory reporting	Data gathering, reconciliation, template population	Data lineage and quality ownership, reviewing automated reconciliations, explaining variances
Compliance advisory / regulatory change	Reading and summarising rules, manual impact assessment	Validating AI-generated horizon scanning and obligation mapping, interpreting regulatory intent
Model risk / validation	Statistical validation of credit and market models	Validating AI/ML and generative AI models, explainability, bias and fairness testing, SS1/23 lifecycle governance
Operational and ICT risk	RCSA facilitation, incident logs	DORA ICT and third-party risk, resilience testing, AI incident management
Internal audit	Cyclical assurance, manual sampling	Data-led and continuous auditing, auditing AI systems and governance, a larger advisory role

Table 4.12 New AI risk and compliance role families: share of public postings

Role family	Share of postings	Top hiring sector
AI Compliance Officer	22%	Financial services
Model Risk Manager	18%	Banking
Responsible AI Lead	14%	Big Tech / SaaS
AI Governance Analyst	12%	Healthcare
AI Risk & Controls Specialist	10%	Insurance
AI Audit Lead	8%	Big 4 / advisory
Privacy & AI Counsel	7%	Legal / tech
AI Policy Manager	5%	Public sector
Other / hybrid	4%	Cross-industry

Source: Global Skill Development Council (2026).

The same report found that AI risk and compliance postings grew 187% year-on-year while general technology hiring contracted by 12%. Financial services accounted for 32% of postings, 68% of postings requested an AI governance certification, and hybrid skill sets (governance language plus model-lifecycle fluency) commanded a 24% pay premium (Global Skill Development Council, 2026). The

dataset is drawn mainly from US-hosted postings and is used as a directional indicator for the UK and EU.

Table 4.13 Skills coded from public role profiles

Category	Skills appearing repeatedly
Regulatory	EU AI Act, DORA, SS1/23 and model risk principles, SM&CR, Consumer Duty, AML rulebooks, data protection; NIST AI RMF; ISO/IEC 42001
Technical	Model risk management, AI/LLM evaluation, explainability tooling, bias and fairness testing, data lineage, privacy-by-design, vendor and third-party AI risk, AI incident response, red-teaming; SQL and Python for analytical roles
Analytical	Risk quantification, interpreting model outputs, root-cause analysis, full-population data analysis
Behavioural	Professional scepticism and effective challenge, translating technical issues for boards and regulators, stakeholder influence, change leadership, continuous learning

4.5.4 Workforce management practices (Objectives iii and iv)

Table 4.14 Evidence on workforce management practices

Evidence	Finding	Source
Skills gaps	46% of professionals report skills gaps within their teams, especially in technology and data knowledge and in technical domain expertise	Thomson Reuters (2025)
Expected impact	80% of professionals expect AI to have a high or transformative impact on their profession within five years, but only 38% expect that change this year	Thomson Reuters (2025)
Governance roles	Organisations that add new roles to their governance and actively invest in AI report more benefits than those that do not	Thomson Reuters (2025)
Internal audit skills	96% agree experienced internal auditors must increase their technology skills; 81% agree entry-level auditors will need an elevated skill set	IIA (2024)
Internal audit role mix	The assurance share of work is expected to fall from 76% to 59%, and the advisory share to rise from 24% to 41%	IIA (2024)
Engagement gap	50% of auditors with 30+ years' experience feel valued, against 21% of those with 0–9 years	IIA (2024)
Agentic AI in KYC/AML	Human effort shifts to exception handling, validation oversight, customer interaction and final decision-making under a human-in-the-loop design	IBM (2026)

Inference. GRC teams are moving from a broad base of junior processors towards a narrower base and a wider middle of experienced reviewers and specialists, a shift from a pyramid to a diamond. Combined with the lower engagement of early-

career professionals, this raises a talent-pipeline question: how will future experts be developed if entry-level processing roles shrink?

4.6 STATISTICAL TESTING OF HYPOTHESES ON PUBLIC DATA

This section reports the tests of hypotheses H5–H10, set out in Table 3.7. The significance level is $\alpha = 0.05$.

4.6.1 AI adoption across EU economies (H5, dataset D1)

Table 4.15 Mean AI adoption and ICT-specialist share across 27 EU countries

Year	AI adoption, 10+ employees (%)	AI adoption, 250+ employees (%)	ICT specialists (% of employment)
2021	7.7	27.5	4.8
2023	8.2	30.2	5.1
2024	14.3	41.2	5.3
2025	20.7	53.6	5.4

Table 4.16 Test results for H5

Test	Statistic	p-value	Effect	n
H5a paired t-test, 250+ employees, 2021→2025	t = 18.99	<0.0001	+26.1 pp; dz = 3.66	27
H5a Wilcoxon signed-rank, 250+ employees, 2021→2025	W = 378	<0.0001	27 of 27 countries up	27
H5a paired t-test, 10+ employees, 2021→2025	t = 10.35	<0.0001	+13.0 pp; dz = 1.99	27
H5a Friedman, 250+ employees, four years	$\chi^2 = 73.31$	<0.0001		27
H5b Pearson, ICT share vs AI adoption (10+), 2025	r = 0.75	<0.0001		27
H5b OLS, AI adoption (10+) on ICT share, 2025	t = 5.70	<0.0001	b = 5.09 pp per 1 pp; R^2 = 0.57	27

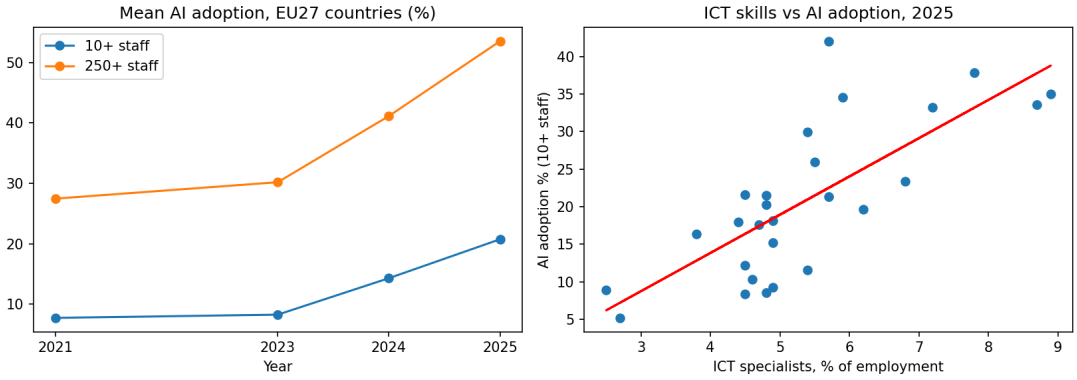


Figure 4.1 AI adoption over time and its relationship with ICT-specialist share (EU27, Eurostat)

Interpretation. H5a and H5b are supported. AI adoption among large enterprises roughly doubled between 2021 and 2025 and rose in every EU country, with the sharpest increase after 2023, when generative AI reached the market. Countries with a larger share of ICT specialists adopt more AI: each additional percentage point of ICT specialists is associated with about five percentage points more AI adoption. This supports the view that AI adoption and technical skills move together. Because the Eurostat survey excludes the financial sector, the result is context for, not direct evidence of, banking practice.

4.6.2 Bank annual reports (H10, dataset D2)

Table 4.17 Mean (median) term density per 10,000 words, 19 paired banks

Measure	FY2022	FY2025
AI terms	0.27 (0.15)	2.87 (2.58)
Reskilling terms	0.21 (0.12)	0.37 (0.30)
AI-governance terms	0.88 (0.82)	1.03 (0.75)
Regulation terms	0.03 (0.00)	0.22 (0.15)
Banks naming an AI governance body	5%	42%

Table 4.18 Test results for H10

Test	Statistic	p-value	Effect	n
H10a AI density, paired t-test	t = 6.61	<0.0001	+2.61; dz = 1.52	19
H10a AI density, Wilcoxon	W = 190	<0.0001	19 of 19 banks up	19
H10a reskilling density, paired t-test	t = 2.09	0.026	+0.16; dz = 0.48	19
H10a reskilling density, Wilcoxon	W = 142	0.030	13 of 19 banks up	19

Test	Statistic	p-value	Effect	n
H10a AI-governance density, paired t-test	t = 1.56	0.068	+0.15; dz = 0.36	19
H10a AI-governance density, Wilcoxon	W = 132	0.072	13 of 19 banks up	19
H10a regulation density, paired t-test	t = 5.42	<0.0001	+0.20; dz = 1.24	19
H10a regulation density, Wilcoxon	W = 171	<0.0001	18 of 19 banks up	19
H10b AI governance body, exact McNemar	–	0.016	7 banks 0→1; 0 banks 1→0	19
H10c AI vs reskilling density, FY2025, Spearman	ρ = −0.16	0.487		22
H10d AI density, UK vs EU, FY2025, Mann–Whitney	U = 87	0.002 (Holm 0.007)	Median 3.96 vs 2.00	6 + 16
H10d reskilling density, UK vs EU	U = 23	0.070 (Holm 0.140)	Median 0.22 vs 0.42	6 + 16
H10d AI-governance density, UK vs EU	U = 94	0.0001 (Holm 0.0004)	Median 2.01 vs 0.50	6 + 16
H10d regulation density, UK vs EU	U = 45	0.858 (Holm 0.858)	Median 0.21 vs 0.15	6 + 16

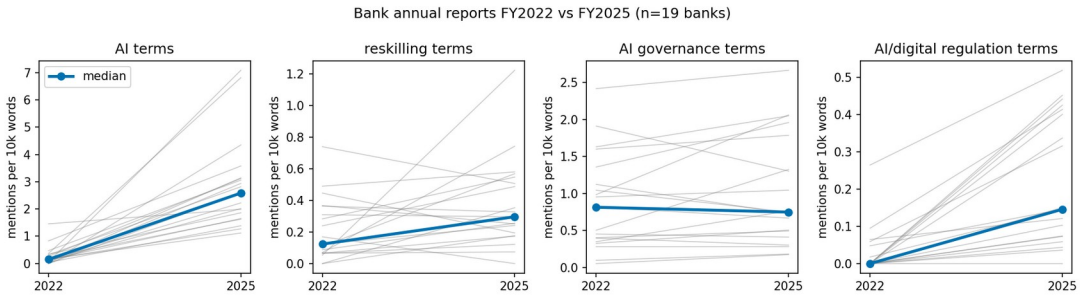


Figure 4.2 Change in AI, reskilling, AI-governance and regulation term density in bank annual reports, FY2022 to FY2025

Interpretation.

- **H10a is supported for AI, reskilling and regulation terms.** AI references rose about tenfold and increased in every bank. References to the AI Act, DORA and SS1/23 went from almost none to routine. Reskilling references rose, but only modestly.
- **H10a is not supported for AI-governance terms** (p ≈ 0.07). Model-risk and oversight language was already common in FY2022 and did not grow much.

- **H10b is supported.** The share of banks naming a dedicated AI governance body rose from 1 in 19 to 8 in 19, and no bank dropped one. Banks are building formal AI governance structures.
- **H10c is not supported.** Banks that talk most about AI do not talk most about reskilling. AI disclosure has grown far faster than reskilling disclosure, and the two are not linked.
- **H10d is partly supported.** UK banks refer to AI and AI governance significantly more than EU banks, which is consistent with the UK's accountability-based regime (SS1/23 and SM&CR). Reskilling and regulation references show no significant regional difference.

4.6.3 Job adverts (H6 and H7, dataset D3)

Of the 163 UK financial-services GRC adverts, 44.2% mention AI or machine learning (95% CI 36.8–51.8%).

Table 4.19 Skill demand in AI-exposed and other adverts (H6)

Skill	AI-exposed (%)	Other (%)	Test	p (Holm)	Odds ratio
Data	51.4	25.3	$\chi^2 = 11.79$	0.002	3.13
Regulatory	40.3	78.0	$\chi^2 = 24.15$	<0.0001	0.19
Cyber	19.4	5.5	$\chi^2 = 7.60$	0.018	4.15
Automation	48.6	13.2	$\chi^2 = 24.58$	<0.0001	6.23
Certification	15.3	12.1	$\chi^2 = 0.35$	1.000	1.31
Governance	52.8	48.4	$\chi^2 = 0.32$	1.000	1.19

Table 4.20 Advertised annual salary by role group (H7)

Comparison	Test	Statistic	p-value	Medians	n
AI-exposed vs other	Mann–Whitney	U = 898.5	0.193	£70,000 vs £59,250	31 vs 52
AI-titled vs traditional	Mann–Whitney	U = 453.5	0.009	£90,000 vs £60,000	8 vs 75

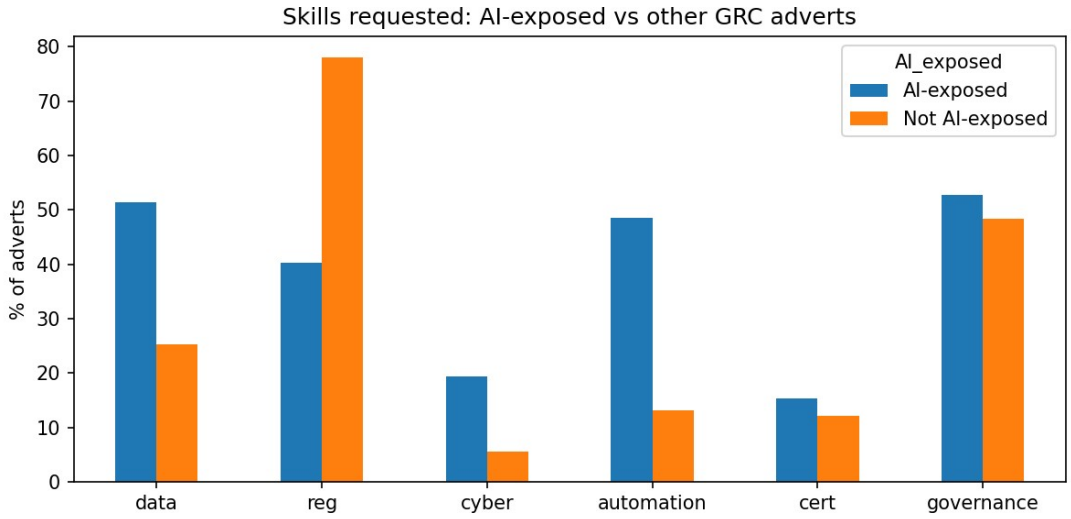


Figure 4.3 Skill demand and advertised salary in AI-exposed and other GRC adverts

Interpretation. H6 is supported for data, regulatory, cyber and automation skills. AI-exposed adverts ask for data skills about three times as often (in odds terms), automation skills six times as often and cyber skills four times as often, but regulatory skills far less often. This is the hybrid-skills gap in practice: employers are hiring technical profiles for AI work, yet practitioners and regulators say that regulatory judgement remains essential. H7 is supported only for roles with AI in the job title, which pay a median of £90,000 against £60,000. That group is very small, so the result is indicative only. For the broader group of AI-exposed adverts, the pay difference is not significant.

4.6.4 Published surveys (H8 and H9, dataset D4)

Table 4.21 Test results for H8 and H9

Test	Statistic	p-value	Effect	n
H8 UK FS firms using AI, 2022→2024 (two-proportion z)	z = 2.41	0.008	58% → 75%	71 / 118
H9 Talent/skills a large or medium constraint (one-sample z vs 50%)	z = 1.49	0.069	57%	118
H9 Experienced auditors need more technology skills (IIA)	z = 189.4	<0.0001	96%	6,506
H9 High or transformative AI impact expected (Thomson Reuters)	z = 35.8	<0.0001	80%	2,275

Interpretation. H8 is supported: the share of UK financial firms using AI rose significantly between the 2022 and 2024 supervisory surveys. H9 is supported for the professional surveys, where large majorities expect AI to transform their profession and agree that experienced staff must increase their technology skills. The firm-level Bank of England and FCA item (57% citing skills as a large or medium constraint) is just short of significance.

4.6.5 Summary of hypothesis results

Table 4.22 Summary of hypothesis results

Hypothesis	Data	Result
H1 Usefulness above midpoint	Questionnaire	Supported descriptively (3.58)
H2 Ease of use above midpoint	Questionnaire	Supported descriptively (4.00)
H3 Roles will change substantially	Questionnaire	Supported descriptively (4.33)
H4 Trust is conditional	Questionnaire	Partly supported
H5a AI adoption increased	D1	Supported ($p < 0.0001$)
H5b ICT skills associated with AI adoption	D1	Supported ($r = 0.75$)
H6 Skill demand differs in AI-exposed adverts	D3	Supported for data, regulatory, cyber and automation skills
H7 AI roles pay more	D3	Partly supported (AI-titled roles only)
H8 UK FS AI adoption rose	D4	Supported ($p = 0.008$)
H9 Skills seen as a constraint by a majority	D4	Partly supported (professional surveys yes; firm-level item $p = 0.069$)
H10a Disclosure density rose	D2	Supported for AI, reskilling and regulation; not for AI governance
H10b More banks name an AI governance body	D2	Supported ($p = 0.016$)
H10c AI and reskilling density correlated	D2	Not supported
H10d UK and EU banks differ	D2	Partly supported (AI and AI governance)

4.7 TRIANGULATION OF EVIDENCE

4.7.1 Practitioner views and industry data

Table 4.23 compares the questionnaire results with the public-data tests for each testable theme.

Table 4.23 Convergence of practitioner views and public data

Theme	Questionnaire (mean, 1–5)	Public data (statistical test)	Converge?
AI use is spreading and roles will change (T1–T3)	Role will change substantially: item 20 = 4.33 (0.58)	AI adoption in large EU firms 27.5% → 53.6% (H5a, $p < 0.0001$); UK FS firms using AI 58% → 75% (H8, $p = 0.008$); bank-report AI references up in 19 of 19 banks (H10a, $p < 0.0001$)	Yes
Data and technical skills more important (T7)	Data skills more important: item 24 = 4.00 (1.00)	AI-exposed adverts ask for data skills 51% vs 25% and automation 49% vs 13% (H6, Holm $p \leq 0.002$)	Yes
Training lags demand (T8/T9)	Training adequate: item 23 = 3.33 (1.53), the lowest skills item	Reskilling references grew far less than AI references (H10a); no link between AI and reskilling intensity (H10c, $\rho = -0.16$); most professionals say technology skills must grow (H9)	Yes
Oversight exists but is not yet embedded (T4–T6)	Accountable people knowledgeable: item 26 = 4.00 (1.00); oversight requirements clear: item 27 = 3.67 (1.15)	AI Act/DORA/SS1/23 references rose (H10a, $p < 0.0001$); banks naming an AI governance body 5% → 42% (H10b, $p = 0.016$); AI-governance wording flat ($p \approx 0.07$)	Partly: structures are growing faster than practice
AI roles carry a pay premium (T7)	–	AI-titled roles £90,000 vs £60,000 median (H7, $p = 0.009$); not for broadly AI-exposed roles	Partly

4.7.2 Testing of propositions

Table 4.24 Triangulated assessment of research propositions

Proposition	Primary evidence	Secondary and statistical evidence	Literature	Result
P1: routine tasks automated first; judgement work augmented	T1, T2, T3; item 20	Job descriptions shift from execution to review; agentic KYC/AML keeps humans for exceptions; H5a, H8, H10a (AI)	Autor (2015); Frey and Osborne (2017)	Supported

Proposition	Primary evidence	Secondary and statistical evidence	Literature	Result
P2: regulation makes competence and oversight compliance conditions	T4, T5, T6; items 26–27	DORA, AI Act, SS1/23, SM&CR and AMLR requirements; H10a (regulation), H10b; H10d (UK higher AI governance)	Buckley et al. (2020)	Supported in texts and structures; partly in practice
P3: a hybrid technical and domain skill profile is required	T7, T9; items 24–25	Skills coded from role profiles; H6 (technical-heavy, regulatory-light AI adverts); H5b; H9; 24% hybrid premium	Cascio and Montealegre (2016)	Supported
P4: loss of institutional knowledge is a material restructuring risk	T8	Knowledge capture treated as a prerequisite for AI deployment; reskilling disclosure lags AI disclosure (H10a, H10c)	Extends Autor (2015)	Supported qualitatively; an emergent contribution

Overall. The statistical results support the qualitative findings. AI adoption and AI disclosure in banks have risen sharply since 2022, and so have regulatory references and formal AI governance structures. Skills and reskilling responses are growing more slowly and are not matched to AI intensity. Practitioners see the same gap in the adequacy of training and in how clearly oversight requirements are applied in practice. This combination makes the case for a structured workforce response, which the next two sections set out.

4.8 GRC ROLE-IMPACT MAP

Drawing on the interview themes, the job-description analysis and the statistical results, Table 4.25 classifies GRC activities and roles by the type of impact AI has on them.

Table 4.25 GRC role-impact map

Category	GRC activities and roles
Automated (substitution)	Data collection and consolidation for reporting; cross-format reconciliation; MI and dashboards; rules-based control monitoring; first-level alert triage; document classification and periodic KYC refresh; execution-focused control testing

Category	GRC activities and roles
Augmented (complementarity)	Alert investigation; KYC/CDD risk judgement; regulatory change impact assessment; control design; model validation; audit planning and fieldwork; risk assessment; regulatory reporting review; compliance advisory
Newly created	AI governance lead; AI compliance officer / responsible-AI lead; AI model risk validator; AI audit lead; AI risk and controls specialist; financial crime data scientist; controls automation engineer / RegTech product owner; ICT third-party and resilience risk manager; human-in-the-loop supervisor of AI agents; knowledge and AI-literacy lead
Human-retained	Interpreting regulatory intent; challenging risk decisions; agreeing remediation; assurance to boards and regulators; final accountability under SM&CR and DORA; ethical and conduct judgements

The map shows that AI changes the composition of GRC work more than its volume. Routine execution is substituted, most professional work is augmented, a new layer of AI governance and assurance roles is created, and final judgement and accountability remain with people.

4.9 WORKFORCE TRANSFORMATION FRAMEWORK

The findings are integrated into a seven-stage workforce transformation framework for GRC and HR leadership (Table 4.26). The stages are sequenced: in particular, knowledge protection (Stage 2) comes before role redesign and any headcount reduction (Stage 3), because the experts whose work is automated are also needed to validate the automation.

Table 4.26 Seven-stage GRC workforce transformation framework

Stage	Management decision	Key actions	Grounding
1. Diagnose	Which activities are exposed, and which people hold critical knowledge?	Activity-level mapping (automated, augmented, retained); AI inventory; identify knowledge-critical roles	P1; T2, T8; H5a, H10a
2. Protect knowledge	How is tacit expertise preserved before change?	Knowledge capture, procedure codification and mentoring; no material reductions until capture is complete	P4; T8

Stage	Management decision	Key actions	Grounding
3. Redesign roles	Which roles are redesigned, reduced or created?	Rewrite job descriptions towards review, challenge and system ownership; create AI governance, validation and assurance roles; protect developmental entry roles	Tables 4.11 and 4.25; H6
4. Build oversight competence	Can accountable people meaningfully challenge AI?	Tiered AI literacy (staff, overseers, model owners, board); competence matrix linked to DORA, the AI Act, SS1/23 and SM&CR; evidence of training	P2; T5; Table 4.10; H10b
5. Reskill and source	Build, buy or borrow?	Build hybrid skills through regulation-by-domain modules and embedded prompts; buy scarce AI-risk specialists; borrow vendor capacity under DORA third-party controls	P3; T7, T9; H6, H9, H10c
6. Secure trust and adoption	Will staff rely on AI appropriately?	Explainability at the point of use; clear override routes; demonstrate usefulness in daily work	H1, H2, H4; T10
7. Lead change and retain talent	How are engagement and commitment sustained?	Transparent communication on role impact; career pathways from processing to investigation, quality assurance and data roles; targeted support for junior staff	H3; T10; Construct G

Three enablers cut across all stages: data quality and lineage, a clear accountability map, and continuous monitoring of both AI performance and workforce indicators such as training completion, override rates and staff turnover.

4.10 DELIVERABLES

Table 4.27 maps the deliverables promised in Section 1.6 to where they are provided in this report.

Table 4.27 Deliverables of the study

Deliverable	Where provided
Analysis of changing GRC activities, roles and skill demand	Sections 4.3.1, 4.5.1, 4.5.3; Tables 4.2, 4.11–4.13
Mapping of regulation to competence and oversight obligations	Section 4.5.2; Table 4.10
Statistical assessment of hypotheses on public data	Section 4.6; Tables 4.15–4.22; Figures 4.1–4.3

Deliverable	Where provided
GRC role-impact map	Section 4.8; Table 4.25
Assessment of skill gaps and reskilling strategies	Sections 4.3.3, 4.4, 4.5.4, 4.6.3
Analysis of employee behavioural responses	Sections 4.3.4 and 4.4; Tables 4.6–4.8
Integrated workforce transformation framework	Section 4.9; Table 4.26
Recommendations for GRC and HR leadership	Chapter 5, Section 5.2

CHAPTER 5

CONCLUSIONS

5.1 SUMMARY OF FINDINGS

The findings are summarised below against each secondary objective.

Objective (i): activities automated or augmented, and role demand.

- Rules-based automation of GRC work is established: data collection and consolidation, reconciliation, management information, rules-based control monitoring and first-level alert triage. AI is spreading quickly but is often still at pilot stage in practitioners' own areas (T1–T3).
- Public data confirm the spread. AI adoption among large EU enterprises roughly doubled from 27.5% in 2021 to 53.6% in 2025, and rose in every country (H5a). The share of UK financial firms using AI rose from 58% to 75% (H8). AI references in bank annual reports rose about tenfold between FY2022 and FY2025, increasing in all 19 paired banks (H10a).
- Execution and testing roles contract first, while review, investigation and AI governance roles grow. P1 is supported.

Objective (ii): regulation, oversight and competence.

- DORA, the EU AI Act, SS1/23 and SM&CR convert competence and human oversight into obligations that must be evidenced.
- References to these instruments in bank reports went from near zero to routine (H10a). The share of banks naming a dedicated AI governance body rose from 5% to 42% (H10b). UK banks discuss AI and AI governance more than EU banks (H10d, partly supported).

- Broader AI-governance wording did not rise significantly, and practitioners rated the clarity of oversight requirements in practice (3.67) below the knowledge of accountable people (4.00). The understanding of AI among accountable senior managers is the critical weak point (T5).
- P2 is supported in regulatory texts and governance structures, and partly in day-to-day practice.

Objective (iii): skill gaps and reskilling.

- A hybrid profile combining data, AI, cyber and automation skills with regulatory judgement is required (T7). AI-exposed adverts ask for data skills about twice as often (51% vs 25%), automation skills nearly four times as often (49% vs 13%) and cyber skills more often (19% vs 6%), but regulatory skills less often (40% vs 78%) (H6). Countries with more ICT specialists adopt more AI (H5b).
- Practitioners rate data and judgement skills as more important (4.00), but training adequacy lowest among skills items (3.33). Reskilling disclosure in bank reports rose only modestly and was not linked to AI intensity (H10c). Banks' workforce response lags their AI ambition.
- A pay premium is visible for AI-titled roles (median £90,000 vs £60,000) but not for broadly AI-exposed roles (H7, partly supported).
- The loss of institutional knowledge held by long-serving experts is the major hidden risk of restructuring (T8). P3 is supported; P4 is supported by qualitative evidence as an emergent contribution.

Objective (iv): employee behavioural responses.

- AI tools are perceived as easy to use (4.00) and useful (3.58), but easier to operate than to understand (item 11: 4.33 vs item 12: 3.33).
- Trust is conditional. Practitioners keep a human check in place even when they rate outputs as reliable, and comfort in relying on AI output without re-performing the work is the more dispersed rating.

- Change is accepted as inevitable (item 20: 4.33), while personal concern about replacement varies widely. Anxiety is concentrated among junior staff, and business change management is described as weak (T10).

Objective (v): framework.

- A seven-stage, sequenced workforce transformation framework was developed (Section 4.9): diagnose, protect knowledge, redesign roles, build oversight competence, reskill and source, secure trust and adoption, and lead change and retain talent. Knowledge protection comes before any restructuring.

5.2 SUGGESTIONS AND RECOMMENDATIONS

5.2.1 For GRC leadership

1. **Map AI exposure at the level of activities, not job titles.** Use the role-impact map (Table 4.25) to classify each team's activities as automated, augmented, newly created or human-retained before changing headcount.
2. **Capture knowledge before reducing headcount.** Identify knowledge-critical roles, then codify procedures, record decision rationales and set up mentoring. Do not make material reductions until this is complete.
3. **Rewrite job descriptions towards review, challenge and system ownership.** Create AI governance, model validation and AI assurance roles. Keep developmental entry-level roles so that future experts can still be trained.
4. **Make human oversight operational.** Define who reviews which AI output, how disagreements are recorded and how outputs can be overridden. Provide explainability at the point of use.

5.2.2 For HR leadership

1. **Close the training gap.** Training adequacy was the lowest-rated skills item, and reskilling disclosure has not kept pace with AI. Build tiered AI-literacy programmes for staff, overseers, model owners and the board, and keep records as compliance evidence.
2. **Develop hybrid skills through domain-specific learning.** Practitioners asked for dedicated learning modules for each regulation within each GRC domain, and for prompts embedded in AI tools.
3. **Use a build, buy and borrow strategy deliberately.** Build hybrid skills internally, buy scarce AI-risk and model-risk specialists, and borrow vendor capacity under DORA third-party controls.
4. **Create visible career pathways.** Show junior staff how processing roles lead to investigation, quality assurance, data and AI governance roles. Communicate role impact early and openly.

5.2.3 For boards and accountable senior managers

1. **Close the understanding gap at the top.** Senior managers accountable under SM&CR and DORA should receive targeted training on AI capabilities, limits and failure modes, so that they can challenge AI-driven outputs rather than simply trust them.
2. **Link AI strategy to workforce strategy.** Every AI deployment plan in GRC should include a matching people plan covering skills, oversight roles, knowledge capture and communication.
3. **Track workforce indicators alongside AI indicators.** Monitor training completion, override rates, staff turnover in affected teams and engagement among junior staff, not only automation savings.

5.3 CONCLUSIONS

This study set out to examine how AI-driven transformation in UK and EU banking GRC is reshaping workforce management decisions, and to develop a practical framework for managing that change. Combining practitioner interviews, a questionnaire, regulatory analysis and statistical tests on four public datasets, it reaches four conclusions.

First, AI in GRC recomposes roles rather than removing them. Routine, rules-based work is absorbed first, while interpretation, challenge and accountability stay with people. This is consistent with Autor's (2015) distinction between substitution and complementarity, now observed in a regulated banking function.

Second, regulation makes workforce capability auditable. DORA, the AI Act, SS1/23 and SM&CR do not prescribe headcount, but they do prescribe who must understand what and who is answerable. Banks are responding with governance structures and regulatory language faster than with day-to-day oversight practice.

Third, the workforce response lags the technology. AI adoption and AI disclosure have risen sharply and consistently, while reskilling has grown far more slowly and is not matched to AI intensity. Practitioners see this gap directly in the adequacy of training.

Fourth, the most serious risk is hidden. When experienced staff leave before their knowledge is captured, the bank loses the very capability it needs to validate and govern AI. This emergent finding extends the automation literature and places knowledge protection at the start of the proposed framework.

The seven-stage framework gives GRC and HR leaders a sequenced, regulation-aware plan. If banks follow it, AI can improve compliance effectiveness without hollowing out the expertise on which compliance depends.

5.4 DIRECTIONS FOR FUTURE RESEARCH

1. **Larger practitioner samples.** A larger, stratified questionnaire across the three lines of defence and both jurisdictions would allow formal tests of H1–H4, reliability analysis and comparisons between practitioner and managerial respondents.
2. **Longitudinal tracking.** Repeating the bank-report and job-advert analysis annually would show whether reskilling disclosure catches up with AI disclosure, and how roles evolve as the AI Act's high-risk obligations take effect.
3. **EU labour-market data.** Extending the job-advert analysis to EU job boards would allow direct UK–EU comparison of skill demand and pay.
4. **Case studies inside banks.** With institutional access, case studies could link AI deployments to actual headcount, training spend, override rates and staff turnover.
5. **Knowledge loss.** The knowledge-loss risk identified in this study merits dedicated research on how GRC expertise can be captured and transferred during automation.
6. **Agentic AI.** As agentic AI begins to handle multi-step KYC and AML work, research is needed on the design of human-in-the-loop supervisory roles and the skills they require.

APPENDIX 1

WRITTEN INTERVIEW: PRACTITIONER RESPONSE FORM

AI-Driven Transformation in Banking GRC: Implications for Workforce Management in UK and EU Banks

About this study. This MBA project examines how the adoption of Artificial Intelligence in the Governance, Risk and Compliance (GRC) functions of UK and EU banks, in regulatory monitoring, screening, risk scoring, control testing and compliance reporting, is changing the work of the people in those functions.

Objective. To examine how AI-driven transformation in GRC functions is reshaping workforce management decisions on role structure, skill requirements, staffing and employee adaptation, and to develop a practical framework that banking management can use to plan GRC workforce transformation while sustaining compliance effectiveness and employee engagement.

This form replaces a 30 to 40 minute interview and takes about 20 minutes. Brief notes or bullet points are entirely acceptable, and you may skip any question.

Confidentiality. Participation is voluntary. Your responses are anonymous. Neither you, nor your employer, nor any client will be identified in my report. Please do not include confidential, proprietary or client-specific information; I am interested in general practice and your professional view.

☐ I consent to my anonymised responses being used for this academic study.

Background (context only, not reported individually)

1. Role / job title:

2. Years of experience in GRC or a GRC-supporting role:
3. Extent of AI or automation use in your area: Not at all / Piloted only / Used in some processes / Used extensively

Section 1: Current use of AI in GRC work

1. Where is AI or automation actually being used today in the risk and compliance work you see, and which of those uses are genuinely in production rather than pilots?
2. Which GRC activities in your area have been most affected so far, and which have proved resistant to automation? What makes the difference?

Section 2: Changes in role content and team structure

3. Thinking about your own role or your team's roles, what tasks take up materially less time than two or three years ago, and what work has replaced them?
4. Have you seen GRC roles disappear, change substantially, or be newly created, and how has the shape of the team changed in headcount or seniority mix? Please describe an example without naming the institution.

Section 3: Regulation, human oversight and accountability

5. How do regulatory expectations on human oversight and accountability, such as DORA, the EU AI Act, PRA SS1/23, or the SM&CR in the UK, affect how AI can actually be used in your area?
6. What does a person need to know or be able to do in order to meaningfully review and challenge an AI-generated risk or compliance output, and are the people who are accountable in that position today?

Section 4: Skills and reskilling

7. Which skills have become more important in GRC work, and where do you see the largest gaps?

8. How is the skills question being handled, through training existing staff, hiring differently, or relying on technology or vendor teams? How effective has that been?

Section 5: Employee response and management of the change

9. How have people in your area responded to AI being introduced into their work, in terms of confidence in the tools, concern about their own role, and willingness to rely on the output? Where output is not trusted or is worked around, what drives that?
10. What has management done well, and what would you do differently, in managing this transition?

Closing

11. Is there anything important about this transition that my questions have not covered?

APPENDIX 2

STRUCTURED QUESTIONNAIRE

Purpose. To test whether the themes emerging from the interviews hold across a wider group of practitioners. Mode: online form. Participation is voluntary and responses are anonymous.

Part 1: Respondent profile

1. Function: Risk / Compliance / Internal Audit / Financial Crime / GRC Technology / HR supporting GRC / Other
2. Level: Analyst or Executive / Manager / Senior Manager / Director or above
3. Years of experience in GRC or a GRC-supporting role: Under 2 / 2–5 / 6–10 / Over 10
4. Primary jurisdiction of your work: UK / EU / Both
5. Type of organisation: Bank / Banking service provider or GCC / Consultancy or vendor serving banks
6. Extent to which AI or automation is used in your area of GRC work: Not at all / Piloted only / Used in some processes / Used extensively

Part 2: Items

All items are rated on a five-point scale: 1 = Strongly disagree, 2 = Disagree, 3 = Neither agree nor disagree, 4 = Agree, 5 = Strongly agree.

A. Perceived usefulness of AI in GRC work (adapted from Davis, 1989)

7. Using AI-enabled tools improves my performance in my GRC work.
8. Using AI-enabled tools increases the volume of work I can complete.

9. AI-enabled tools improve the quality and consistency of risk and compliance outputs in my area.
10. Overall, I find AI-enabled tools useful in my job.

B. Perceived ease of use (adapted from Davis, 1989)

11. I find the AI-enabled tools in my area easy to use.
12. It is easy for me to understand what the AI-enabled tools in my area are doing.
13. Learning to work with these tools did not require excessive effort.

C. Trust in AI output (adapted from Glikson & Woolley, 2020)

14. I can understand how the AI-enabled tools in my area arrive at their outputs.
15. The outputs produced by these tools are reliable.
16. I am confident I could explain and defend an AI-assisted risk or compliance decision to a regulator or auditor.
17. I am comfortable relying on AI-generated output without independently re-performing the work.
18. When I disagree with an AI-generated output, there is an effective way to challenge or override it.

D. Perceived replaceability (adapted from Brougham & Haar, 2018)

19. I am concerned that a significant part of my current role could be performed by AI or automation.
20. I believe my role will change substantially over the next three years because of AI and automation.
21. I am concerned about my long-term career prospects in GRC because of AI and automation.

E. Skills and reskilling

22. My current skills are sufficient for the way GRC work is being performed in my organisation.
23. My organisation has provided adequate training to work effectively with AI-enabled GRC tools.
24. Data and analytical skills have become more important in my role.
25. Judgement, interpretation and regulatory reasoning have become more important in my role than routine processing.

F. Human oversight and accountability

26. The people accountable for AI-assisted GRC decisions in my area have the knowledge required to supervise them effectively.
27. Regulatory requirements for human oversight are clearly reflected in how AI is used in my area.

G. Organisational response and outcomes (adapted from Brougham & Haar, 2018)

28. Management has communicated clearly how AI adoption will affect roles in my function.
29. I feel positive about the effect of AI adoption on my work.
30. I would consider leaving my current role because of the way AI and automation are changing it.
31. I feel committed to my organisation.

Part 3: Open questions (optional)

32. Which parts of your GRC work do you think should not be automated, and why?
33. What single change would most help GRC staff adapt to AI-enabled ways of working?

APPENDIX 3

BANK ANNUAL REPORT SAMPLE AND CODING DICTIONARY

A3.1 Banks in the D2 sample

All reports are public annual reports (or universal registration documents) for FY2022 and FY2025, taken from each bank's official investor-relations website. Analysis tables in Chapter 4 use codes rather than names, so that no individual bank is singled out.

Table A 3.1 Banks in the D2 sample

Coverage	Banks
Both FY2022 and FY2025 (paired tests)	ABN AMRO (EU), BBVA (EU), Banco Santander (EU), Bank of Ireland (EU), Barclays (UK), Danske Bank (EU), Deutsche Bank (EU), Erste Group (EU), Groupe BPCE (EU), HSBC Holdings (UK), Intesa Sanpaolo (EU), KBC Group (EU), NatWest Group (UK), Nationwide Building Society (UK), Nordea (EU), Santander UK (UK), Société Générale (EU), Standard Chartered (UK), UniCredit (EU)
FY2025 only (FY2025 cross-section)	Handelsbanken (EU), ING Group (EU), SEB (EU)
FY2022 only (not used in tests)	Rabobank (EU)
Not coded (no full English report available, or automated access blocked)	AIB Group, BNP Paribas, CaixaBank, Commerzbank, Crédit Agricole, Lloyds Banking Group

Nationwide Building Society's financial year ends in April; its reports for the years ending April 2023 and April 2026 are mapped to FY2022 and FY2025.

A3.2 D2 coding dictionary

Matching is case-insensitive on the full extracted text. Densities are counts per 10,000 words.

Table A 3.2 D2 coding dictionary

Measure	Terms counted
AI	artificial intelligence; machine learning; generative AI; GenAI; large language model(s); LLM(s); "AI" as a separate word
Reskilling	reskill / re-skill*; upskill / up-skill*; retrain*; digital skill(s); AI literacy; data literacy; skill(s) gap; skill(s) development
AI governance	AI governance; responsible AI; AI ethics; ethical AI; human oversight; human-in-the-loop; explainab*; model risk
Regulation	AI Act; Artificial Intelligence Act; DORA; digital operational resilience; SS1/23
AI governance body (0/1)	chief AI / artificial intelligence / data / data and AI officer; AI council, committee, board, office or centre of excellence; responsible AI office, council, committee, lead, team or framework

A3.3 D3 skill dictionary

Table A 3.3 D3 skill flags

Skill flag	Terms in advert text
AI-exposed	AI; artificial intelligence; machine learning; LLM(s); generative; GenAI
Data	SQL; Python; R programming; data analysis / analytics; Power BI; Tableau; data science
Regulatory	FCA; PRA; regulat*; DORA; AI Act; SM&CR; Consumer Duty; Basel; GDPR
Cyber	cyber; information security; ISO 27001; NIST
Automation	automat*; RPA
Certification	CIPP; AIGP; CAMS; CISA; CISSP; CRISC; ACCA; CFA; ICA; IAPP; ISO 42001; FRM; PRM; CIA
Governance	governance; oversight; accountab*

AI-titled roles are those with AI, artificial intelligence, machine learning, ML, GenAI, generative or LLM in the job title.

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